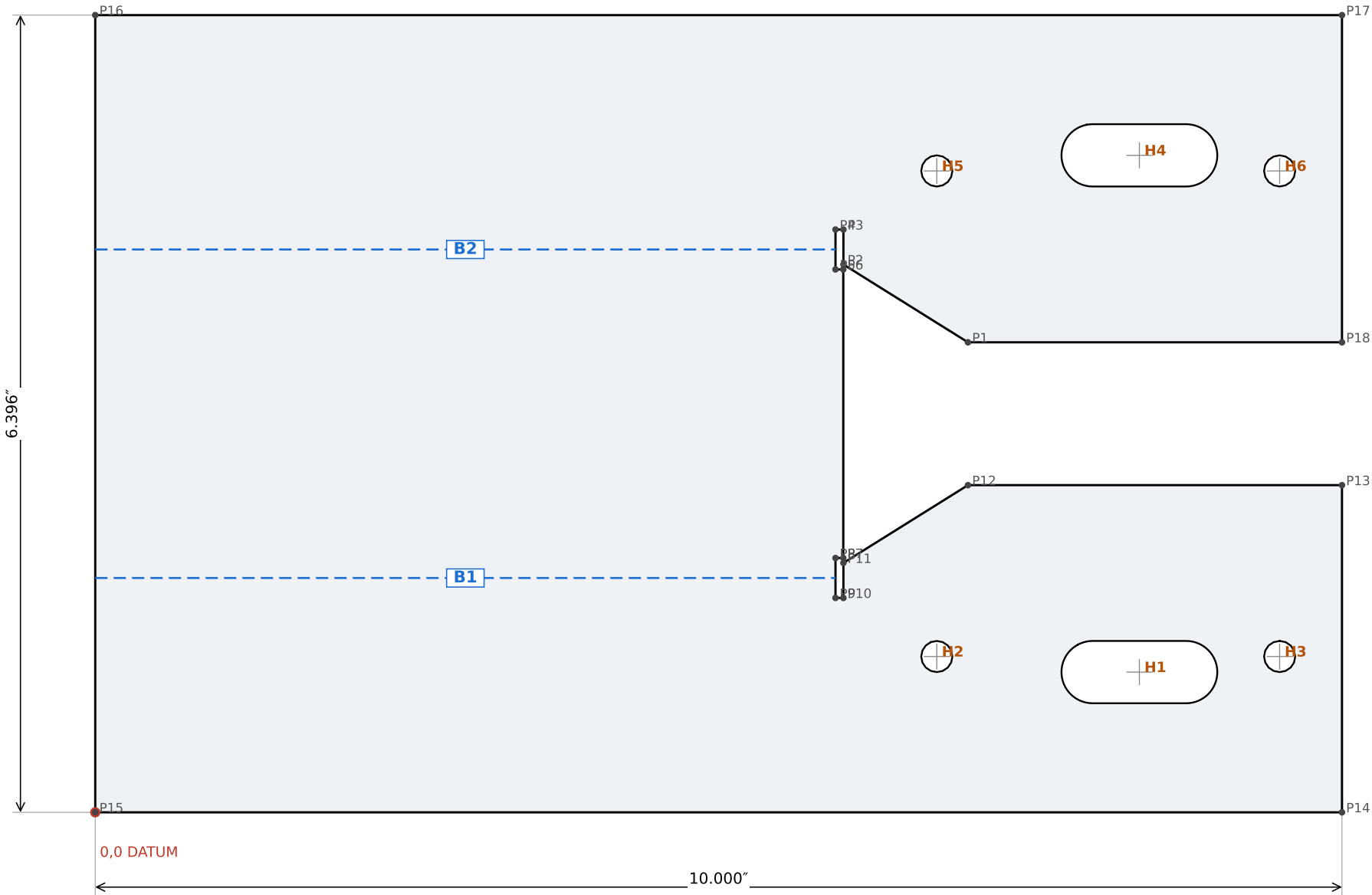


FF-1 — blank 1 of 1

Channel half with ears — flat pattern (2 bends 90°, web on top) · blank 10.000" × 6.396" · HR steel sheet 0.138" · 2 per vehicle



HOLES (centre X, Y from datum)				
H1	slot 1.250 × 0.500"	@0°	X 8.375	Y 1.125
H2	Ø0.250"		X 6.750	Y 1.250
H3	Ø0.250"		X 9.500	Y 1.250
H4	slot 1.250 × 0.500"	@0°	X 8.375	Y 5.271
H5	Ø0.250"		X 6.750	Y 5.146
H6	Ø0.250"		X 9.500	Y 5.146
BENDS (bend centre line, from datum) IR = t = 0.138"				
B1	(1)/(3) 90°		(0.000, 1.880) → (5.938, 1.880)	
B2	(2)/(4) 90°		(0.000, 4.516) → (5.938, 4.516)	

OUTLINE POINTS (X, Y from datum)		
P1	7.000	3.771
P2	6.001	4.395
P3	6.001	4.676
P4	5.938	4.676
P5	5.938	4.356
P6	6.000	4.356
P7	6.000	2.040
P8	5.938	2.040
P9	5.938	1.720
P10	6.001	1.720
P11	6.001	2.001
P12	7.000	2.625
P13	10.000	2.625
P14	10.000	0.000
P15	0.000	0.000
P16	0.000	6.396
P17	10.000	6.396
P18	10.000	3.771

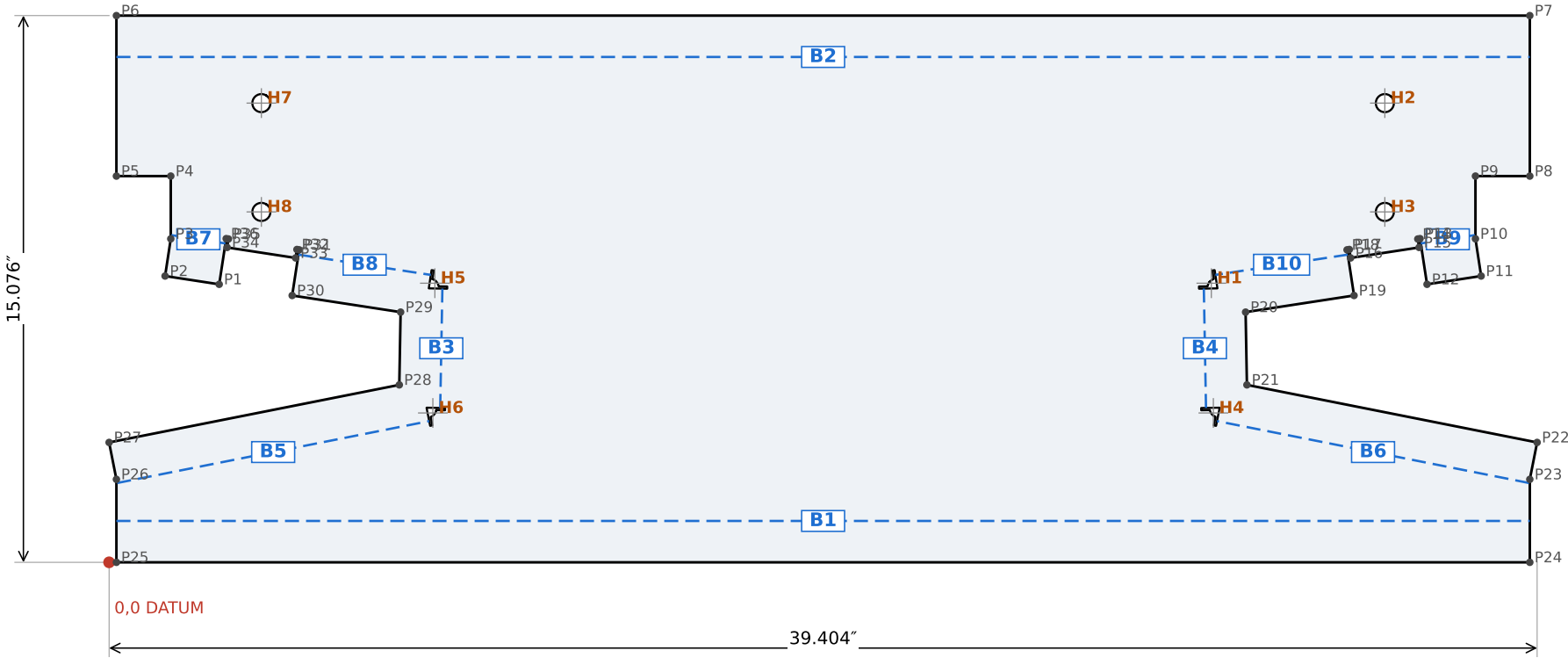
NOTES

1. Cut from the DXF file (1:1, inches). This sheet is for checking.
2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
3. Flat pattern already includes bend allowance (K = 0.44, R = t).
4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * web on top, level with D; bend the walls 90° down on the dashed lines; the ears reach past the bend line (no web there); 1/16" relief slit at the bend end; weld the two halves in the middle at ~175° (ends ¼" out)

FF-1 — blank 1 of 1	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Channel half with ears — flat pattern (2 bends 90°, web on top)	QTY: 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

FF-3 — blank 1 of 4

FF3-1 top plate — flat pattern, 10 bends 90° down · blank 39.404” × 15.076” · HR steel sheet 0.118” · 1 per vehicle



HOLES (centre X, Y from datum)

H1	slot	0.509 × 0.505"	@1°	X	30.414	Y	7.696
H2	Ø0.500"			X	35.202	Y	12.663
H3	Ø0.500"			X	35.202	Y	9.663
H4	slot	0.511 × 0.499"	@1°	X	30.470	Y	4.118
H5	slot	0.509 × 0.505"	@-1°	X	8.989	Y	7.696
H6	slot	0.511 × 0.499"	@-1°	X	8.934	Y	4.118
H7	Ø0.500"			X	4.202	Y	12.663
H8	Ø0.500"			X	4.202	Y	9.663

BENDS (bend centre line, from datum) IR = t = 0.118"

B1	(3)	90° ↓	(0.202, 1.141) → (39.202, 1.141)
B2	(4)	90° ↓	(0.202, 13.935) → (39.202, 13.935)
B3	(5)	90° ↓	(9.137, 4.258) → (9.196, 7.547)
B4	(6)	90° ↓	(30.267, 4.258) → (30.207, 7.547)
B5	(7)	90° ↓	(0.222, 2.187) → (8.833, 3.894)
B6	(8)	90° ↓	(39.182, 2.187) → (30.571, 3.894)
B7	(9)	90° ↓	(1.717, 9.022) → (3.210, 8.793)
B8	(10)	90° ↓	(5.224, 8.485) → (8.878, 7.925)
B9	(11)	90° ↓	(37.686, 9.022) → (36.193, 8.793)
B10	(12)	90° ↓	(34.179, 8.485) → (30.526, 7.925)

OUTLINE POINTS (X, Y from datum)

P1	3.038	7.666
P2	1.545	7.895
P3	1.702	8.921
P4	1.702	10.651
P5	0.202	10.651
P6	0.202	15.076
P7	39.202	15.076
P8	39.202	10.651
P9	37.702	10.651
P10	37.702	8.921
P11	37.859	7.895
P12	36.366	7.666
P13	36.173	8.925
P14	36.111	8.916
P15	36.147	8.683
P16	34.257	8.393
P17	34.221	8.627
P18	34.159	8.617
P19	34.352	7.358
P20	31.360	6.899
P21	31.396	4.893
P22	39.404	3.306
P23	39.202	2.288
P24	39.202	0.000
P25	0.202	0.000
P26	0.202	2.288
P27	0.000	3.306
P28	8.008	4.893
P29	8.044	6.899
P30	5.052	7.358
P31	5.245	8.617
P32	5.183	8.627
P33	5.147	8.393
P34	3.257	8.683
P35	3.292	8.916
P36	3.231	8.925

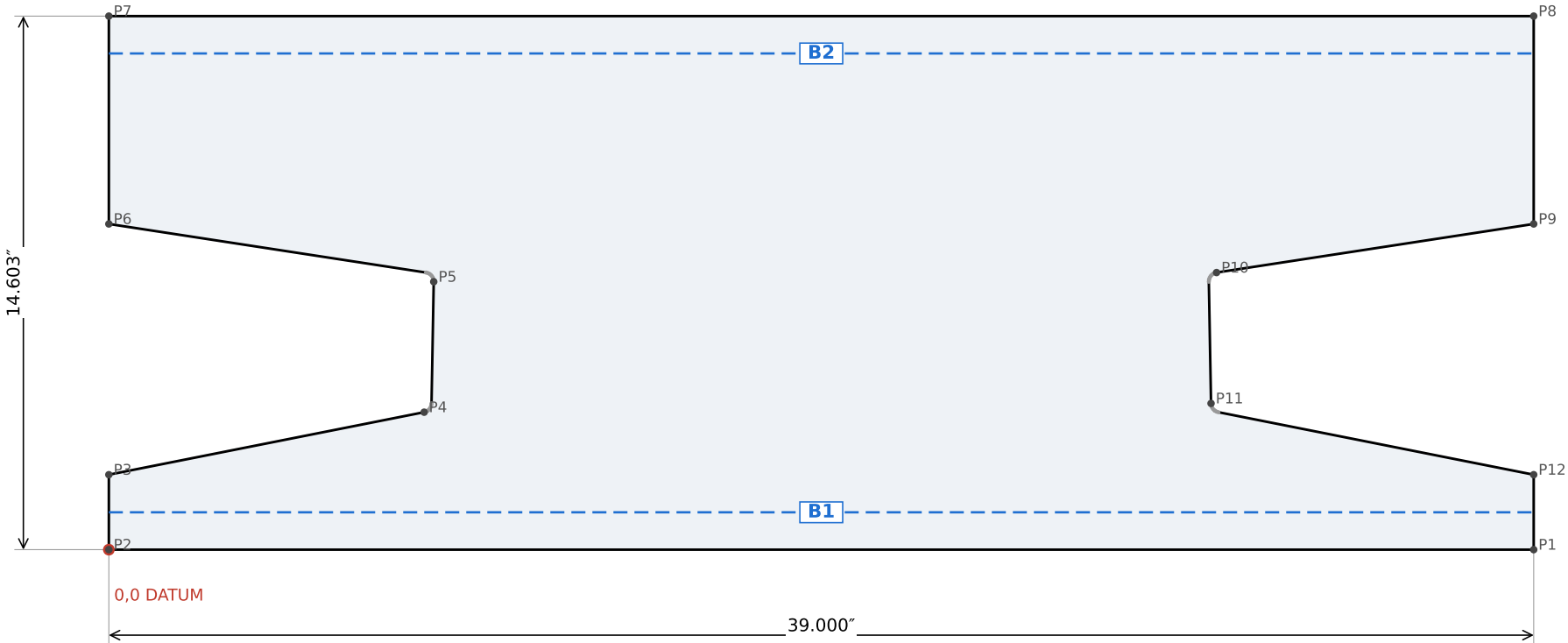
NOTES

1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 2. Material: HR steel sheet 0.118" (t). Inside bend radius = t.
 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * rims 1½" bent down on the long sides and around the whole notch; leg ends open; in the 4 inner notch corners the rims are mitred — a gap stays in the corner; 1/16" reliefs at the bend ends; the 4 Ø½ holes are cut right away

FF-3 — blank 1 of 4	HR steel sheet 0.118"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
FF3-1 top plate — flat pattern, 10 bends 90° down	QTY: 1 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 4

FF-3 — blank 2 of 4

FF3-2 lower plate — flat pattern, 2 bends 90° up · blank 39.000″ × 14.603″ · HR steel sheet 0.118″ · 1 per vehicle



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.118″ (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * rims on the long sides bent up 1″ (to the FF3-1 underside), set in by t inside the FF3-1 rims; 1/16″ reliefs at the bend ends

HOLES (centre X, Y from datum)

none

BENDS (bend centre line, from datum) IR = t = 0.118″

B1 (1) 90° (0.000, 1.022) → (39.000, 1.022)
B2 (2) 90° (0.000, 13.581) → (39.000, 13.581)

OUTLINE POINTS (X, Y from datum)

P1 39.000 0.000
P2 0.000 0.000
P3 0.000 2.052
P4 8.630 3.762
P5 8.892 7.333
P6 0.000 8.914
P7 0.000 14.603
P8 39.000 14.603
P9 39.000 8.914
P10 30.320 7.585
P11 30.168 4.003
P12 39.000 2.052

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

8.677 3.776
8.720 3.799
8.739 3.813
8.774 3.847
8.801 3.887
8.820 3.932
8.830 3.979
8.832 4.003

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

8.889 7.381
8.883 7.404
8.866 7.449
8.841 7.489
8.808 7.524
8.770 7.553
8.726 7.573
8.680 7.585

FREE-FORM CURVE 3 (grey dots, follow DXF) — points X, Y

30.273 7.573
30.230 7.553
30.192 7.524
30.159 7.489
30.134 7.449
30.117 7.404
30.111 7.381
30.108 7.333

FREE-FORM CURVE 4 (grey dots, follow DXF) — points X, Y

30.170 3.979
30.180 3.932
30.199 3.887
30.226 3.847
30.261 3.813
30.280 3.799
30.323 3.776
30.369 3.762

FF-3 — blank 2 of 4	HR steel sheet 0.118″	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
FF3-2 lower plate — flat pattern, 2 bends 90° up	QTY: 1 per vehicle	cut ±0.010″ · bends ±1° · IR = t	SHEET 2 / 4

FF-3 — blank 3 of 4

Passage wall 1⅛ × 4.69" (instead of the tube) · blank 4.694" × 1.125" · HR steel sheet 0.118" · 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	4.694	0.000
P2	4.694	1.125
P3	0.000	1.125
P4	0.000	0.000



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.118" (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

FF-3 — blank 3 of 4	HR steel sheet 0.118"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Passage wall 1⅛ × 4.69" (instead of the tube)	QTY: 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 4

FF-3 — blank 4 of 4

Passage wall 1⅛ × 5.02” (instead of the tube) · blank 5.021” × 1.125” · HR steel sheet 0.118” · 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	5.021	0.000
P2	5.021	1.125
P3	0.000	1.125
P4	0.000	0.000



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.118” (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

FF-3 — blank 4 of 4	HR steel sheet 0.118”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Passage wall 1⅛ × 5.02” (instead of the tube)	QTY: 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 4 / 4

FF-4 — cut list (tubes)

• Square tube 1 ½” — cut 39”

Stock: Square tube 1 ½” × 1 ½”, wall 0.197” · cut length 39.000” · QTY 1 per vehicle

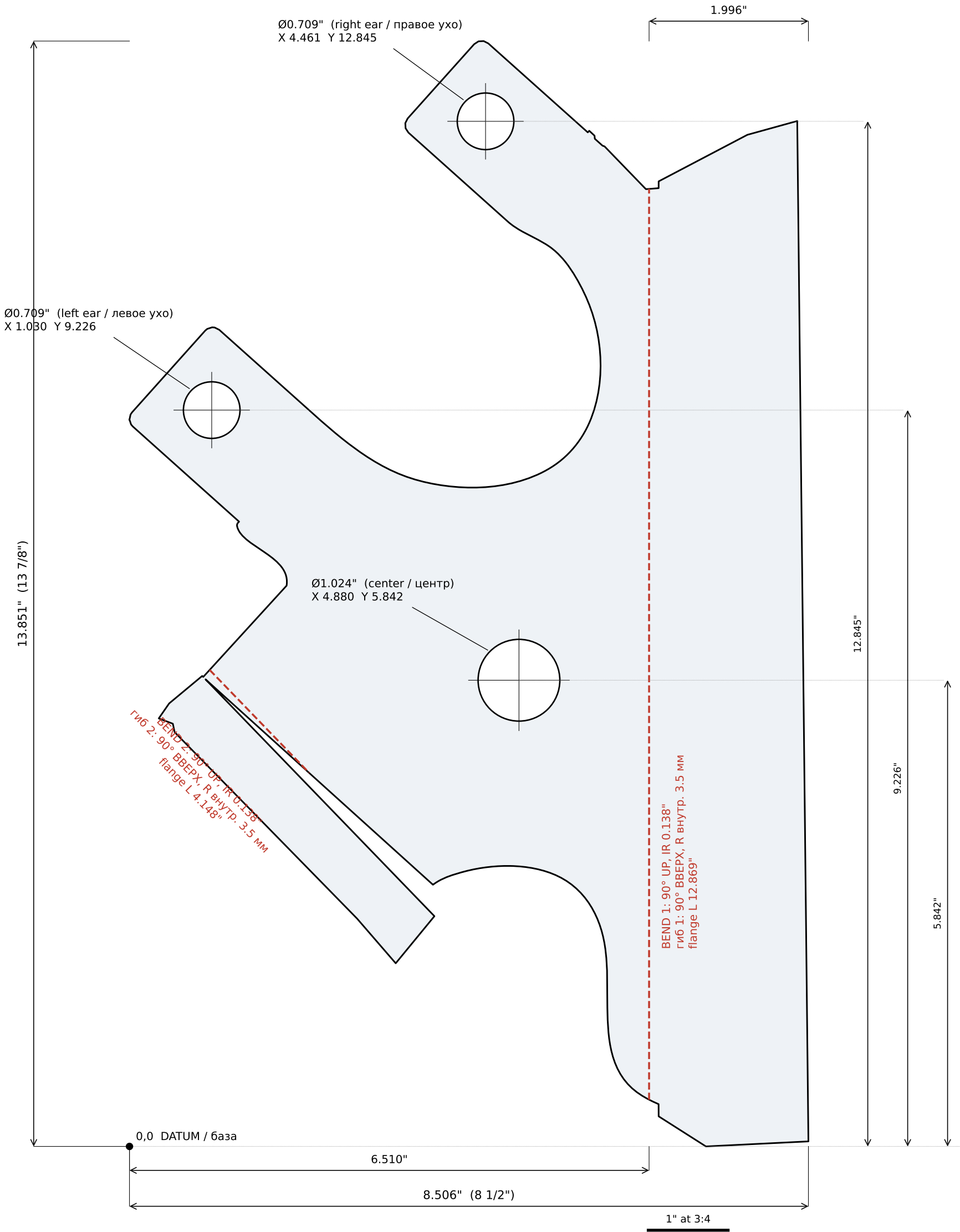
wall 0.197”

Cut square, ±1/32”. Deburr. Lengths in inches.

FF-4 — tubes		inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Square tube 1 ½” — cut 39”	QTY: 1 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 1

FR-1 — FLAT PATTERN / РАЗВЕРТКА (scale 3:4 / масштаб 3:4)

blank 8.506" × 13.851" · 3.5 mm (0.138") plate · 2 bends 90° up · IR 0.138"



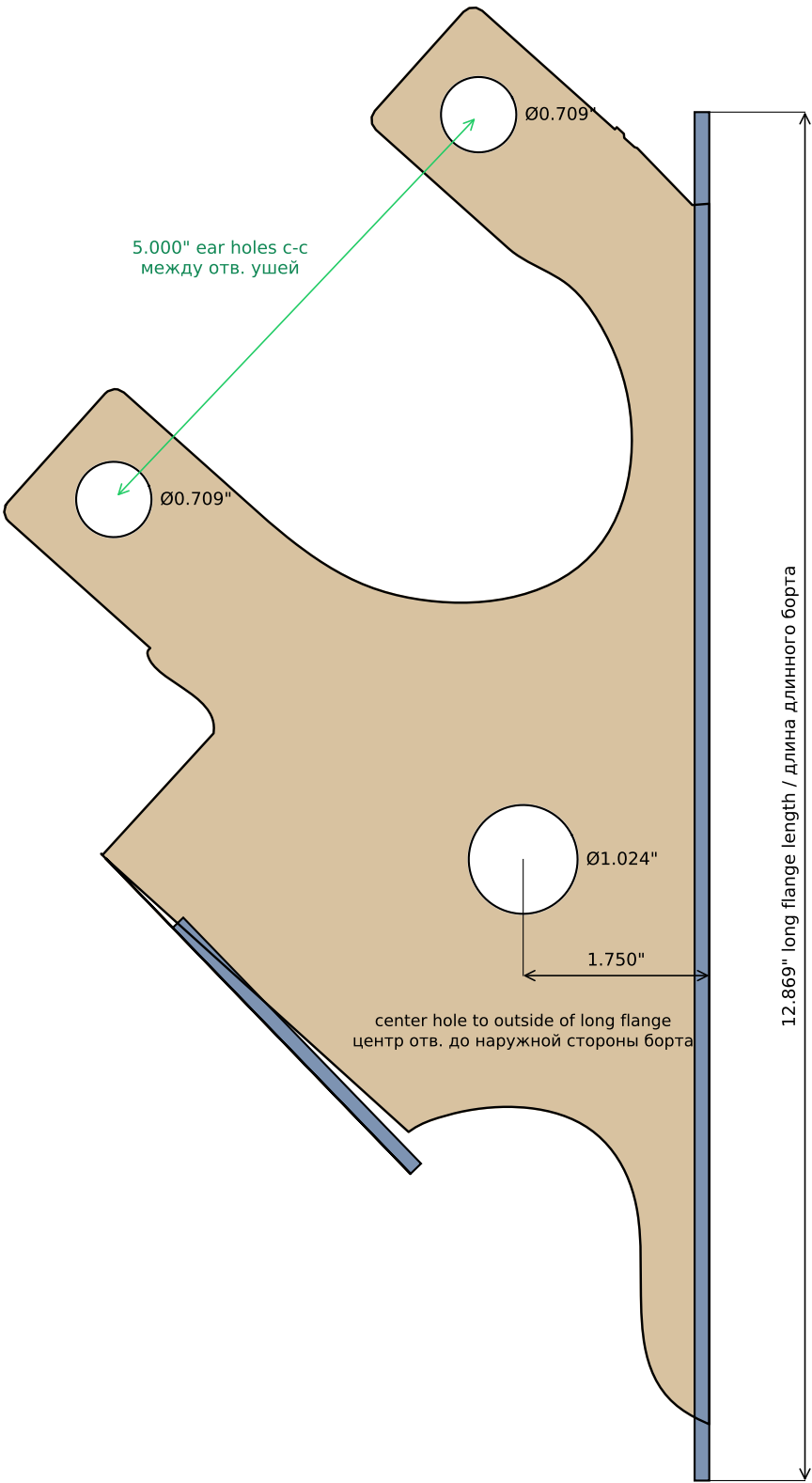
NOTES / ПРИМЕЧАНИЯ

- 1. Cut from the DXF file FR-1_flat_3-5mm.dxf (1:1, inches). This sheet is for checking. Резать по DXF (1:1, дюймы); чертёж — для проверки.
- 2. Both bends 90° UP to the SAME side (toward the viewer), inside radius 0.138" (R = t = 3.5 mm). Обагиба 90° в одну сторону (к себе), R внутр. = 3.5 мм.
- 3. Flat pattern computed with K = 0.44, bend allowance 0.312", bend deduction 0.240" per bend.
If your tooling/K differs, keep the FINISHED sizes on sheet 2 and adjust the blank. Если ваш инструмент другой — держите готовые размеры (лист 2).
- 4. Red dashed lines = bend CENTER lines. Красный пунктир = средняя линиягиба.
- 5. Holes are laser cut in the flat (no drilling). Отверстия режутся лазером сразу.
- 6. QTY 2: FR-1 as shown + FL-1 = SAME blank flipped over (bends to the other side = mirror part).
2 шт: FR-1 как на чертеже + FL-1 = та же заготовка, перевернуть (гнуть в другую сторону = зеркальная деталь).
- 7. Deburr all edges. Снять заусенцы. Curved edges: smooth arcs/splines from the DXF.

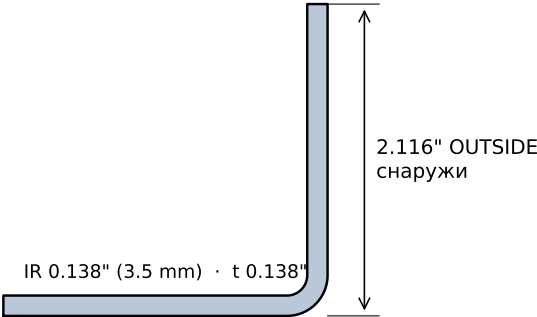
FR-1 · side plate / боковая пластина	A36 HR steel 3.5 mm (0.138")	inch / дюймы	Sienna WAV · 2026-10-02
FLAT PATTERN 3:4 / РАЗВЕРТКА 3:4	QTY 2 (FR-1 + FL-1 mirror)	cut ±0.010" · bend ±1°	SHEET 1 / 2

FR-1 — FINISHED PART / ГОТОВАЯ ДЕТАЛЬ

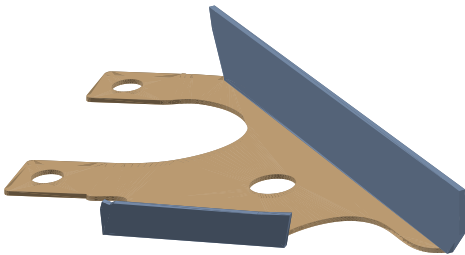
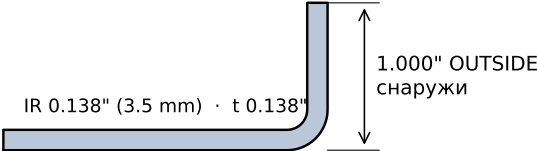
TOP VIEW (flanges up, toward viewer) / ВИД СВЕРХУ (борта вверх, к себе)



BEND 1 · long flange / длинный борт



BEND 2 · slant flange / косой борт



ISOMETRIC / ИЗОМЕТРИЯ

BEND TABLE / ТАБЛИЦА ГИБОВ

#	edge / кромка	angle	dir / направл.	IR	flange length	finished outside
1	long edge / длинная	90°	UP / вверх	0.138"	12.869"	2.116"
2	slant edge / косая	90°	UP / вверх	0.138"	4.148"	1.000"

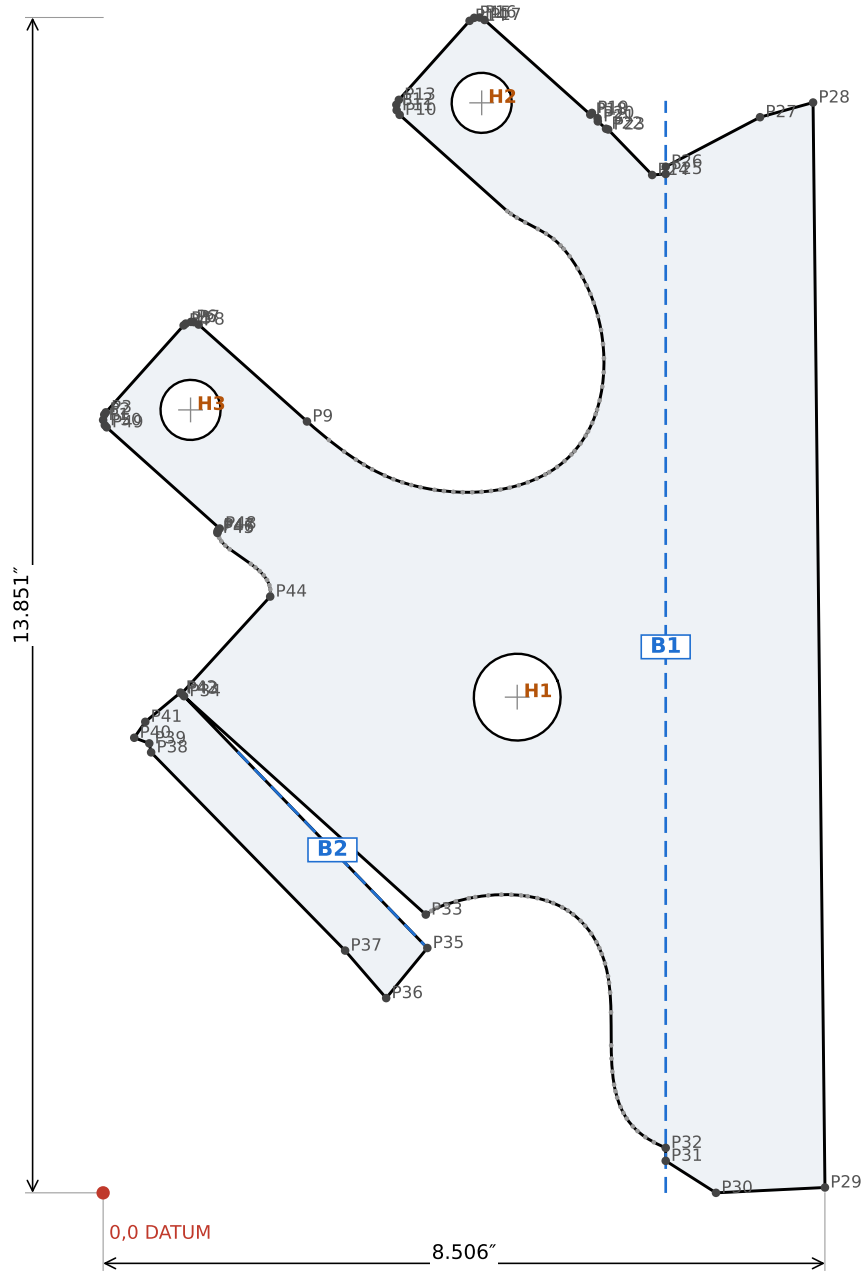
Order: any (bends do not touch). Порядок гибов любой – они не пересекаются.
Both flanges on the SAME side. Оба борта в одну сторону.
FL-1 (mirror): same blank, bend both flanges to the OTHER side. FL-1: та же заготовка, гнуть в другую сторону.

- CHECK AFTER BENDING / ПРОВЕРИТЬ ПОСЛЕ ГИБКИ:
- long flange outside height 2.116" (±1/32")
 - center hole Ø1.024" (26 mm) is 1.750" from the outside of the long flange
 - ear holes Ø0.709" (18 mm) both, are 5.000" apart (c-c) – they bolt to FR-2
 - slant flange outside height 1.000" (±1/32")

FR-1 · side plate / боковая пластина	A36 HR steel 3.5 mm (0.138")	inch / дюймы	Sienna WAV · 2026-10-02
FINISHED PART, BEND TABLE / ГОТОВАЯ ДЕТАЛЬ, ГИБЫ	QTY 2 (FR-1 + FL-1 mirror)	cut ±0.010" · bend ±1°	SHEET 2 / 2

FR-1 — blank 1 of 1

Plate with ears — flat pattern (2 bends 90°) · blank 8.506″ × 13.851″ · HR steel sheet 0.138″ · 1 per side · FR + FL (mirror) = 2 per vehicle



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138″ (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * bend 90° up on dashed lines

HOLES (centre X, Y from datum)

H1	Ø1.024″	X 4.880	Y 5.842
H2	Ø0.709″	X 4.461	Y 12.845
H3	Ø0.709″	X 1.030	Y 9.226

BENDS (bend centre line, from datum) IR = t = 0.138″

B1	(1) 90°	(6.630, -0.001) → (6.630, 12.869)
B2	(2) 90°	(1.584, 5.199) → (3.821, 2.885)

OUTLINE POINTS (X, Y from datum)

P1	0.000	9.108
P2	0.013	9.171
P3	0.032	9.198
P4	0.949	10.222
P5	0.974	10.244
P6	1.035	10.264
P7	1.068	10.261
P8	1.125	10.232
P9	2.402	9.090
P10	3.495	12.704
P11	3.459	12.758
P12	3.456	12.823
P13	3.485	12.880
P14	4.319	13.812
P15	4.373	13.847
P16	4.437	13.851
P17	4.495	13.822
P18	5.741	12.706
P19	5.760	12.727
P20	5.827	12.665
P21	5.830	12.627
P22	5.927	12.540
P23	5.952	12.531
P24	6.470	11.995
P25	6.629	12.007
P26	6.630	12.092
P27	7.741	12.676
P28	8.366	12.849
P29	8.506	0.063
P30	7.222	0.000
P31	6.630	0.377
P32	6.630	0.530
P33	3.802	3.279
P34	0.954	5.853
P35	3.821	2.885
P36	3.336	2.295
P37	2.852	2.855
P38	0.566	5.191
P39	0.544	5.297
P40	0.367	5.363
P41	0.497	5.551
P42	0.911	5.895
P43	0.924	5.883
P44	1.969	7.027
P45	1.347	7.777
P46	1.348	7.793
P47	1.354	7.808
P48	1.374	7.829
P49	0.041	9.021
P50	0.020	9.046

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.661	8.872
2.840	8.736
2.978	8.640

FR-1 — blank 1 of 1	HR steel sheet 0.138″	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate with ears — flat pattern (2 bends 90°)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010″ · bends ±1° · IR = t	SHEET 1 / 1

3.073	8.581
3.170	8.527
3.318	8.453
3.421	8.410
3.525	8.373
3.632	8.341
3.740	8.315
3.905	8.285
4.016	8.271
4.127	8.261
4.238	8.256
4.350	8.256
4.461	8.261
4.572	8.271
4.682	8.288
4.791	8.310
4.899	8.339
5.005	8.374
5.109	8.416
5.209	8.465
5.258	8.493
5.351	8.554
5.438	8.624
5.519	8.701
5.556	8.743
5.626	8.830
5.687	8.924
5.715	8.972
5.763	9.073
5.804	9.177
5.822	9.230

5.851	9.338
5.883	9.502
5.899	9.668
5.902	9.835
5.891	10.003
5.868	10.168
5.831	10.331
5.782	10.490
5.742	10.594
5.696	10.695
5.646	10.796
5.591	10.894
5.498	11.036
5.428	11.123
5.391	11.164
5.309	11.239
5.220	11.303
5.123	11.359
4.973	11.436
4.876	11.491
4.806	11.536
4.762	11.570

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

6.490	0.596
6.389	0.657
6.297	0.728
6.242	0.783
6.171	0.875
6.131	0.942
6.097	1.012
6.057	1.122

FR-1 — blank 1 of 1	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate with ears — flat pattern (2 bends 90°)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

6.037	1.197
6.022	1.273
6.001	1.427
5.991	1.583
5.987	1.738
5.985	2.127
5.974	2.360
5.962	2.475
5.950	2.552
5.915	2.704
5.877	2.814
5.847	2.886
5.812	2.955
5.752	3.055
5.681	3.148
5.628	3.204
5.540	3.281
5.476	3.326
5.409	3.366
5.304	3.415
5.231	3.442
5.118	3.473
5.004	3.495
4.926	3.505

4.810	3.513
4.694	3.515
4.538	3.505
4.384	3.485
4.308	3.471
4.157	3.434
4.045	3.400
3.936	3.358
3.867	3.323

FREE-FORM CURVE 3 (grey dots, follow DXF) — points X, Y

1.972	7.061
1.970	7.111
1.959	7.161
1.931	7.222
1.892	7.277
1.846	7.326
1.795	7.370
1.741	7.411
1.491	7.585
1.440	7.628
1.394	7.678
1.367	7.720
1.353	7.752

FR-1 — blank 1 of 1	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate with ears — flat pattern (2 bends 90°)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

FR-2 (welded version) — blank 1 of 4

Top (2 slots) · blank 6.750” × 3.000” · HR steel sheet 0.138” · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)			
H1	slot	0.749 × 0.625”	@0° X 5.875 Y 1.125
H2	slot	0.709 × 0.472”	@0° X 0.875 Y 1.250
OUTLINE POINTS (X, Y from datum)			
P1	6.750	0.000	
P2	0.000	0.000	
P3	0.000	3.000	
P4	6.750	3.000	



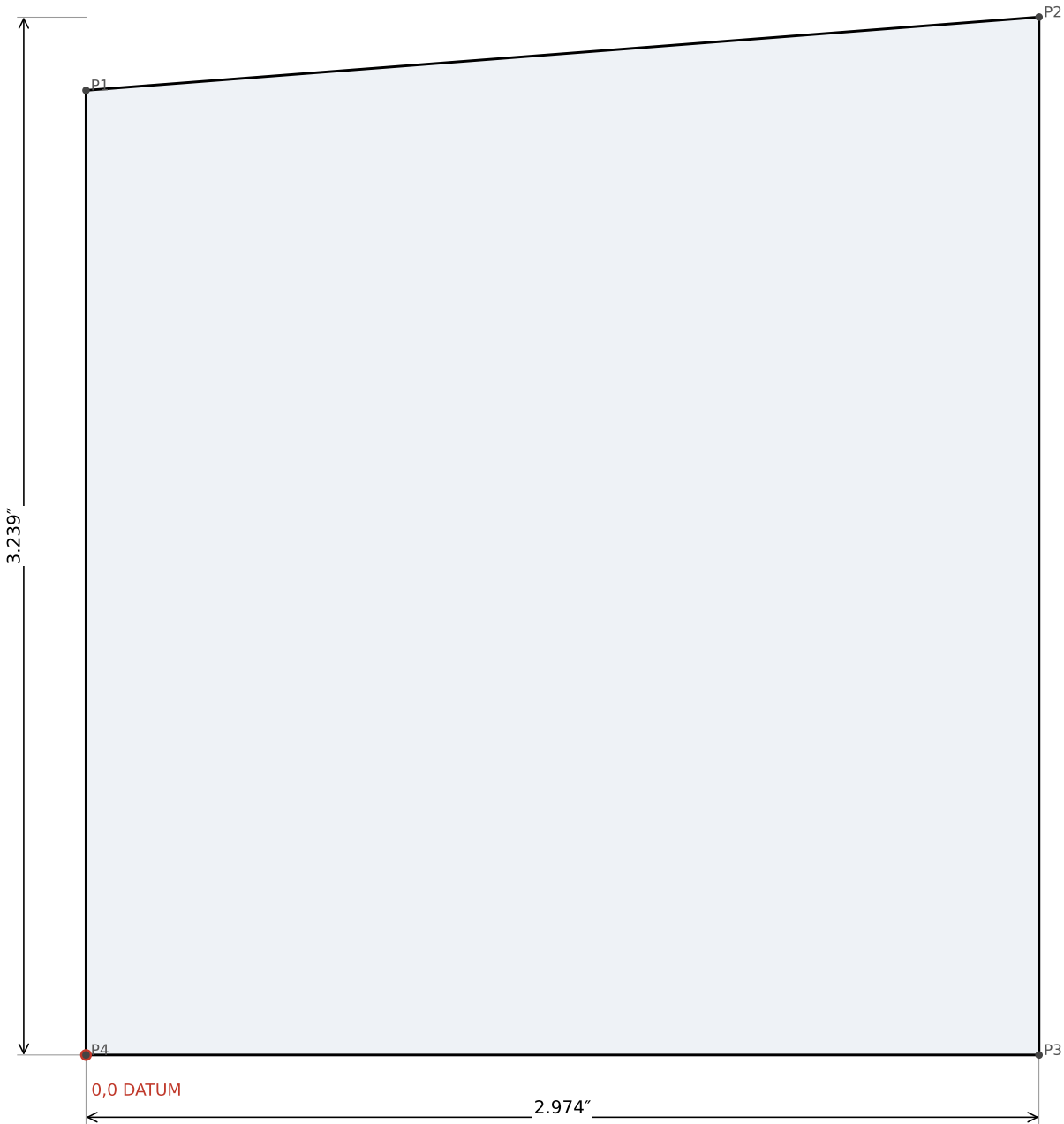
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138” (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-2 (welded version) — blank 1 of 4	HR steel sheet 0.138”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Top (2 slots)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 4

FR-2 (welded version) — blank 2 of 4

Leg wall · blank 2.974" × 3.239" · HR steel sheet 0.138" · 4 per side · FR + FL (mirror) = 8 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	3.011
P2	2.974	3.239
P3	2.974	0.000
P4	0.000	0.000

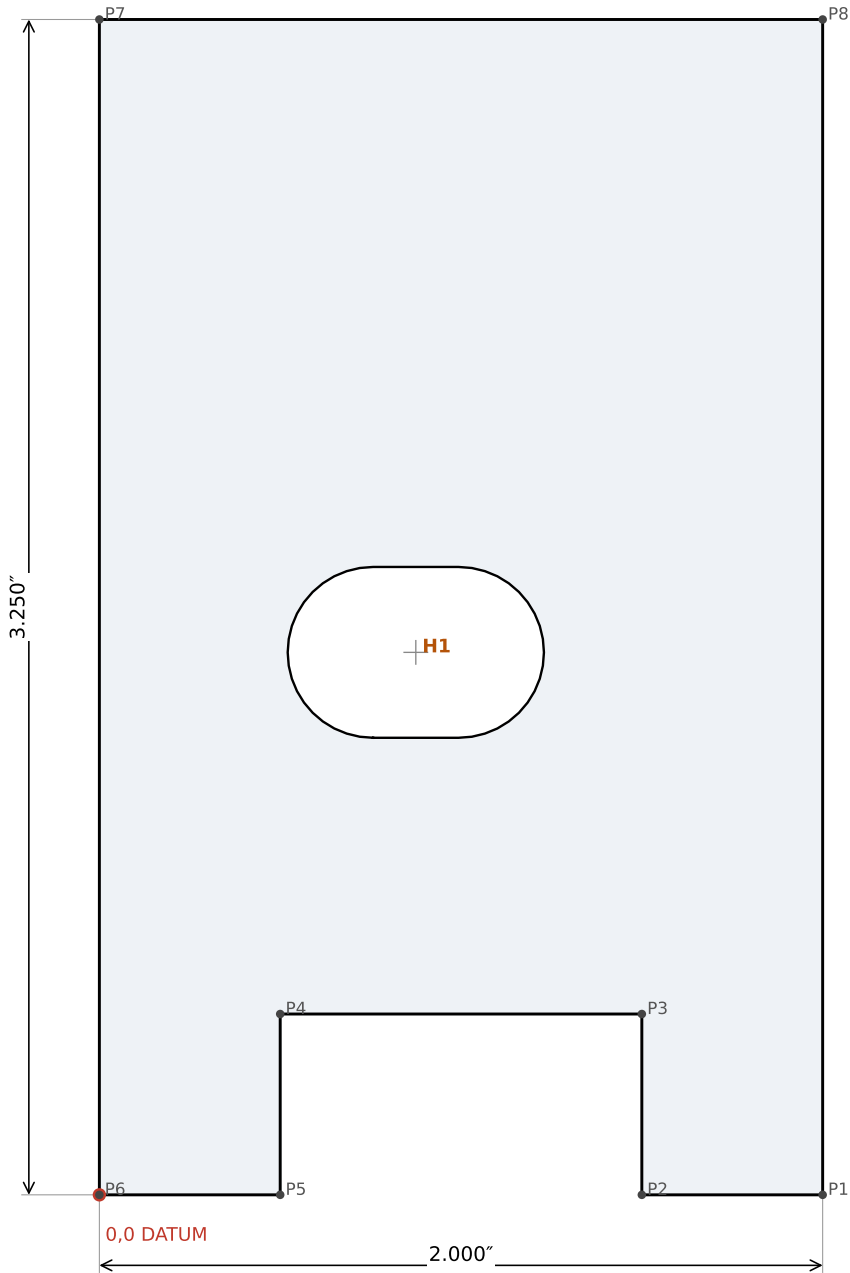
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-2 (welded version) — blank 2 of 4	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Leg wall	QTY: 4 per side · FR + FL (mirror) = 8 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 4

FR-2 (welded version) — blank 3 of 4

Leg bottom, left (slot 0.709×0.472) · blank 2.000" × 3.250" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)		
H1	slot 0.709 × 0.472" @0°	X 0.875 Y 1.500
OUTLINE POINTS (X, Y from datum)		
P1	2.000	0.000
P2	1.500	0.000
P3	1.500	0.500
P4	0.500	0.500
P5	0.500	0.000
P6	0.000	0.000
P7	0.000	3.250
P8	2.000	3.250

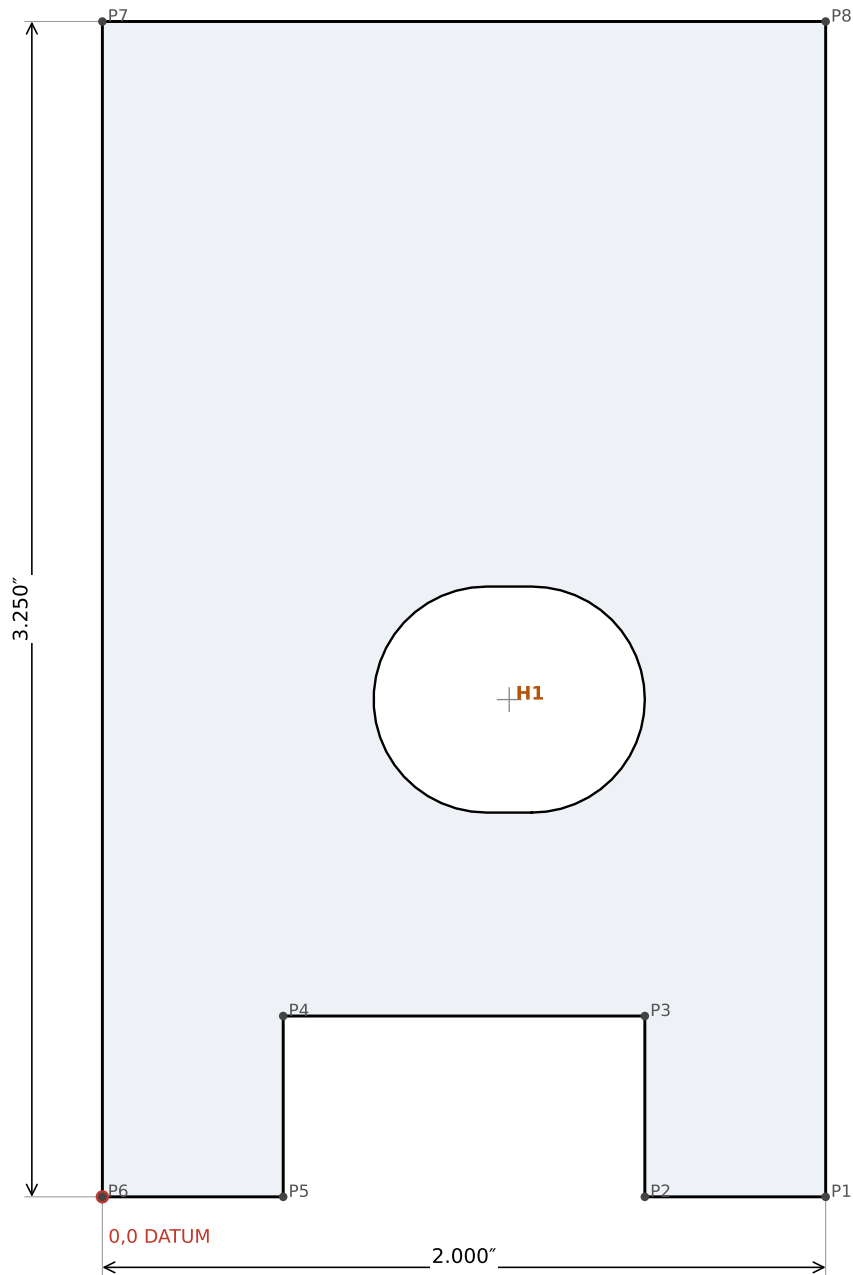
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-2 (welded version) — blank 3 of 4	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Leg bottom, left (slot 0.709×0.472)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 4

FR-2 (welded version) — blank 4 of 4

Leg bottom, right (slot 3/4×5/8) · blank 2.000″ × 3.250″ · HR steel sheet 0.138″ · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)		
H1	slot 0.749 × 0.625″ @0°	X 1.125 Y 1.375
OUTLINE POINTS (X, Y from datum)		
P1	2.000	0.000
P2	1.500	0.000
P3	1.500	0.500
P4	0.500	0.500
P5	0.500	0.000
P6	0.000	0.000
P7	0.000	3.250
P8	2.000	3.250

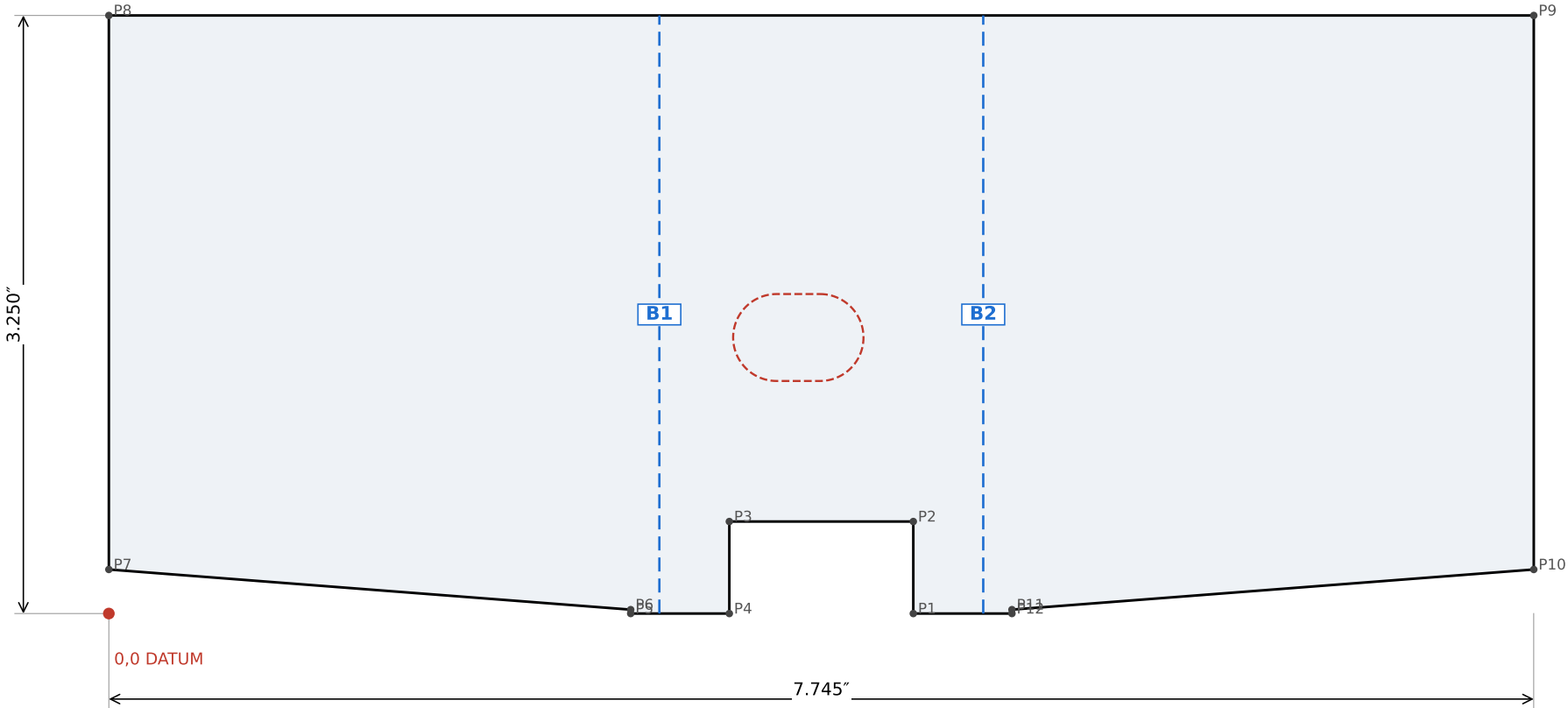
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138″ (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-2 (welded version) — blank 4 of 4	HR steel sheet 0.138″	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Leg bottom, right (slot 3/4×5/8)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010″ · bends ±1° · IR = t	SHEET 4 / 4

FR-2 (bent version) — blank 1 of 3

Leg 1 (U: 2 walls + bottom) · blank 7.745" × 3.250" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

DRILL AFTER WELDING (red dashed — do NOT laser cut)

D1 slot 0.709 × 0.472" X 3.748 Y 1.500

BENDS (bend centre line, from datum) IR = t = 0.138"

B1 (1) 90° (2.992, 0.000) → (2.992, 3.250)

B2 (2) 90° (4.753, 0.000) → (4.753, 3.250)

OUTLINE POINTS (X, Y from datum)

P1	4.373	0.000
P2	4.373	0.500
P3	3.373	0.500
P4	3.373	0.000
P5	2.837	0.000
P6	2.837	0.021
P7	0.000	0.239
P8	0.000	3.250
P9	7.745	3.250
P10	7.745	0.239
P11	4.909	0.021
P12	4.909	0.000

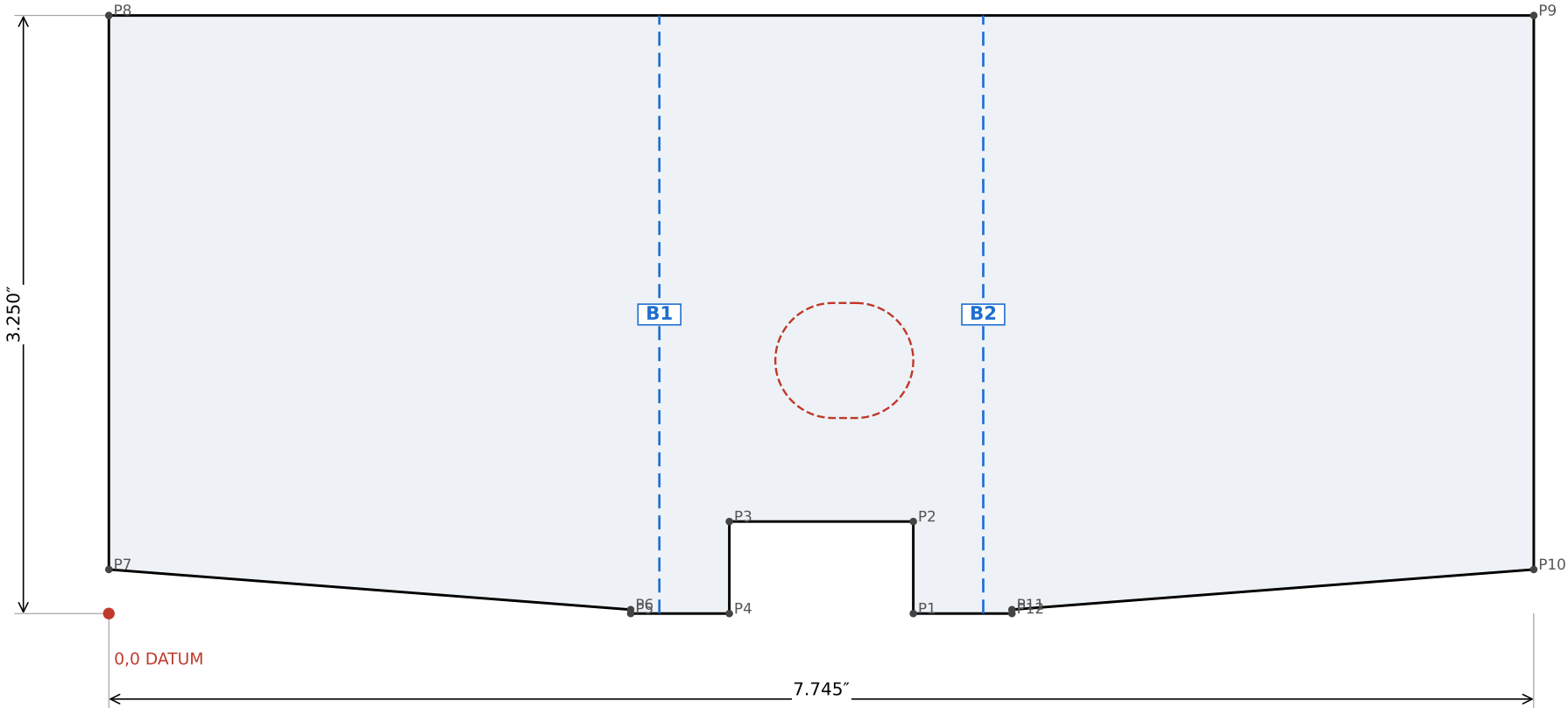
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).
- * 90° bends, R = t; ½" notch at the front. Do NOT cut the 0.709×0.472 oval — drill it after welding through the top oval (centre 0.875" from the outer wall, 1¼" from front edge A)

FR-2 (bent version) — blank 1 of 3	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Leg 1 (U: 2 walls + bottom)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 3

FR-2 (bent version) — blank 2 of 3

Leg 2 (U: 2 walls + bottom) · blank 7.745" × 3.250" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

DRILL AFTER WELDING (red dashed — do NOT laser cut)

D1 slot 0.750 × 0.625" X 3.998 Y 1.375

BENDS (bend centre line, from datum) IR = t = 0.138"

B1 (3) 90° (2.992, 0.000) → (2.992, 3.250)

B2 (4) 90° (4.753, 0.000) → (4.753, 3.250)

OUTLINE POINTS (X, Y from datum)

P1	4.373	0.000
P2	4.373	0.500
P3	3.373	0.500
P4	3.373	0.000
P5	2.837	0.000
P6	2.837	0.021
P7	0.000	0.239
P8	0.000	3.250
P9	7.745	3.250
P10	7.745	0.239
P11	4.909	0.021
P12	4.909	0.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * 90° bends, R = t. Do NOT cut the ¾×⅝ oval — drill it after welding through the top oval (centre 0.875" from the outer wall, 1⅙" from the front edge)

FR-2 (bent version) — blank 2 of 3	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Leg 2 (U: 2 walls + bottom)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 3

FR-2 (bent version) — blank 3 of 3

Top 6 ¾×3 (2 ovals) · blank 6.750″ × 3.000″ · HR steel sheet 0.138″ · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)			
H1	slot 0.709 × 0.472″	@0°	X 0.875 Y 1.250
H2	slot 0.750 × 0.625″	@0°	X 5.875 Y 1.125
OUTLINE POINTS (X, Y from datum)			
P1	0.000	3.000	
P2	6.750	3.000	
P3	6.750	0.000	
P4	0.000	0.000	



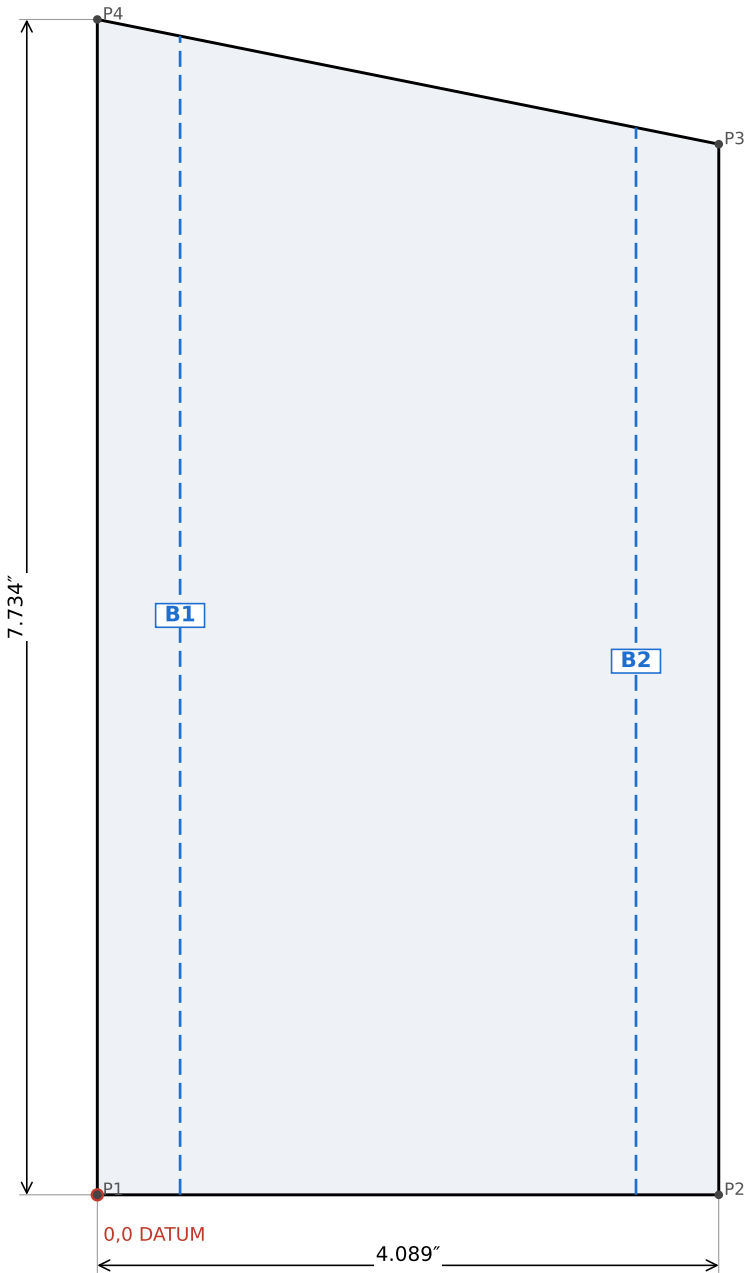
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138″ (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * flat, no bends

FR-2 (bent version) — blank 3 of 3	HR steel sheet 0.138″	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Top 6 ¾×3 (2 ovals)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010″ · bends ±1° · IR = t	SHEET 3 / 3

FR-3 — blank 1 of 1

Channel — flat pattern (2 bends 90°) · blank 4.089” × 7.734” · HR steel sheet 0.118” · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

BENDS (bend centre line, from datum) IR = t = 0.118”

B1 (1) 90° (0.545, 0.000) → (0.545, 7.625)
B2 (2) 90° (3.545, 0.000) → (3.545, 7.023)

OUTLINE POINTS (X, Y from datum)

P1 0.000 0.000
P2 4.089 0.000
P3 4.089 6.914
P4 0.000 7.734

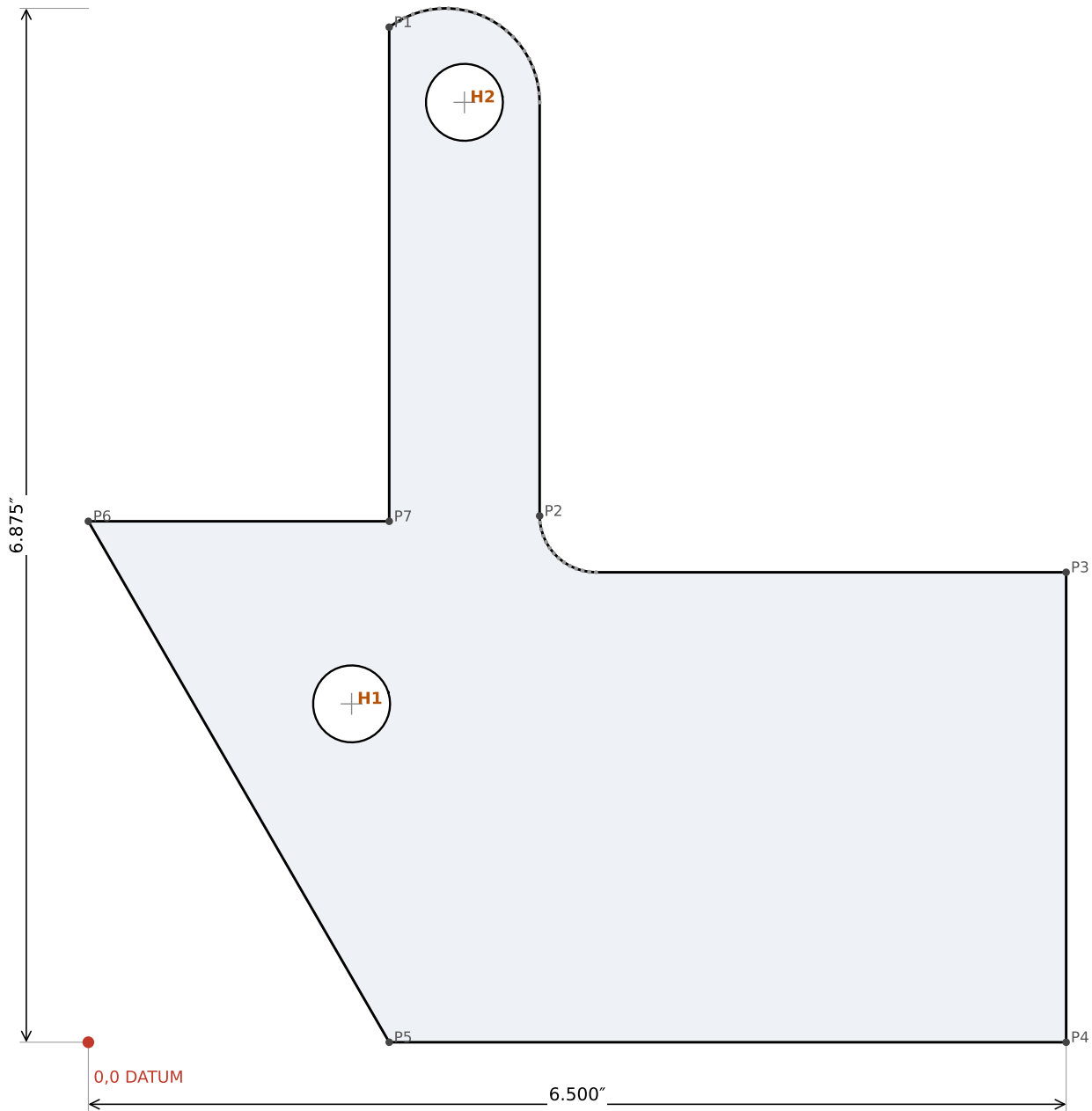
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.118” (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * bend 90° on dashed lines, 3/4 flanges inward

FR-3 — blank 1 of 1	HR steel sheet 0.118”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Channel — flat pattern (2 bends 90°)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 1

FR-4 (welded version) — blank 1 of 8

Side plate, right (hole Ø0.512) · blank 6.500" × 6.875" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.512"	X 1.750	Y 2.250
H2	Ø0.512"	X 2.500	Y 6.250

OUTLINE POINTS (X, Y from datum)

P1	2.000	6.750
P2	3.000	3.500
P3	6.500	3.125
P4	6.500	0.000
P5	2.000	0.000
P6	0.000	3.464
P7	2.000	3.464

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.050	6.784
2.103	6.812
2.158	6.836
2.215	6.854
2.274	6.867
2.334	6.874
2.394	6.875
2.454	6.870
2.513	6.859
2.571	6.843
2.627	6.822
2.681	6.795
2.732	6.763
2.780	6.726
2.824	6.685
2.863	6.640
2.899	6.591
2.929	6.539
2.954	6.485
2.974	6.428
2.988	6.370
2.997	6.310
3.000	6.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.003	3.455
3.011	3.410
3.024	3.367
3.043	3.326
3.066	3.287
3.094	3.251
3.126	3.219
3.162	3.191
3.201	3.168
3.242	3.149
3.285	3.136
3.330	3.128
3.375	3.125

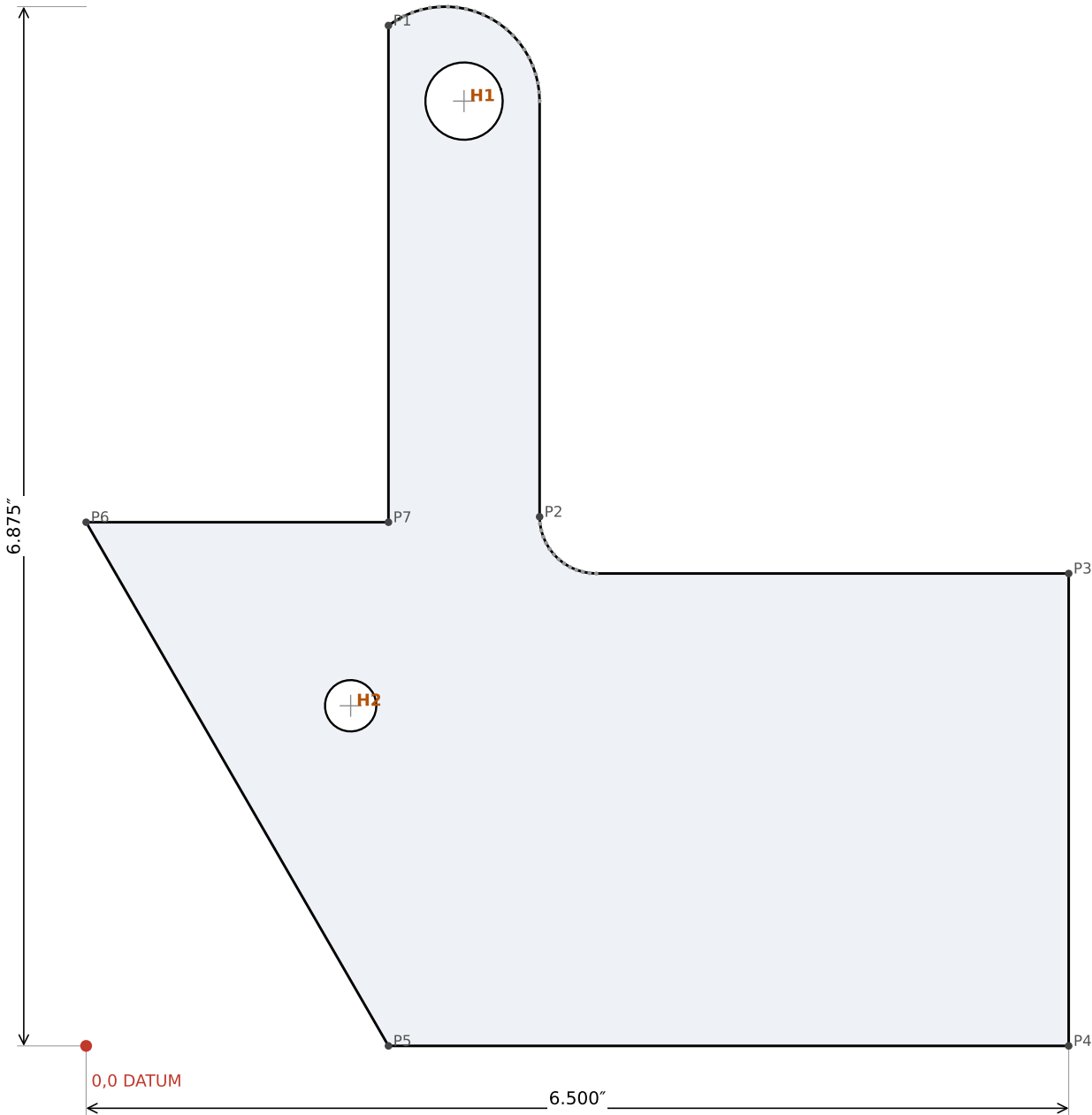
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-4 (welded version) — blank 1 of 8	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate, right (hole Ø0.512)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 8

FR-4 (welded version) — blank 2 of 8

Side plate, left (hole Ø0.34) · blank 6.500" × 6.875" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.512"	X 2.500	Y 6.250
H2	Ø0.340"	X 1.750	Y 2.250

OUTLINE POINTS (X, Y from datum)

P1	2.000	6.750
P2	3.000	3.500
P3	6.500	3.125
P4	6.500	0.000
P5	2.000	0.000
P6	0.000	3.464
P7	2.000	3.464

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.050	6.784
2.103	6.812
2.158	6.836
2.215	6.854
2.274	6.867
2.334	6.874
2.394	6.875
2.454	6.870
2.513	6.859
2.571	6.843
2.627	6.822
2.681	6.795
2.732	6.763
2.780	6.726
2.824	6.685
2.863	6.640
2.899	6.591
2.929	6.539
2.954	6.485
2.974	6.428
2.988	6.370
2.997	6.310
3.000	6.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.003	3.455
3.011	3.410
3.024	3.367
3.043	3.326
3.066	3.287
3.094	3.251
3.126	3.219
3.162	3.191
3.201	3.168
3.242	3.149
3.285	3.136
3.330	3.128
3.375	3.125

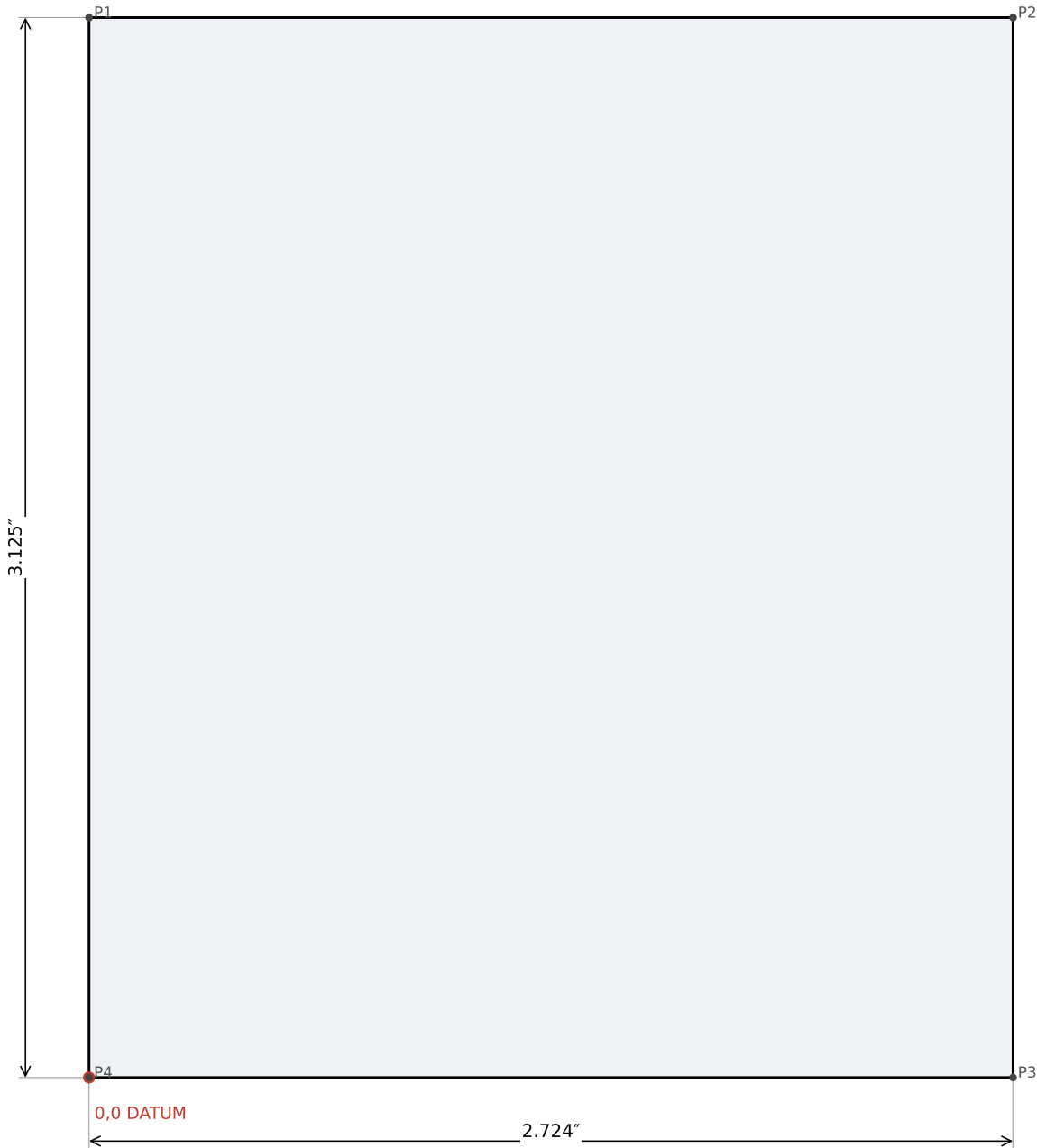
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-4 (welded version) — blank 2 of 8	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate, left (hole Ø0.34)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 8

FR-4 (welded version) — blank 3 of 8

Front plate · blank 2.724" × 3.125" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	3.125
P2	2.724	3.125
P3	2.724	0.000
P4	0.000	0.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-4 (welded version) — blank 3 of 8	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Front plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 8

FR-4 (welded version) — blank 4 of 8

Slanted plate · blank 4.000" × 2.724" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	2.724
P3	4.000	2.724
P4	4.000	0.000

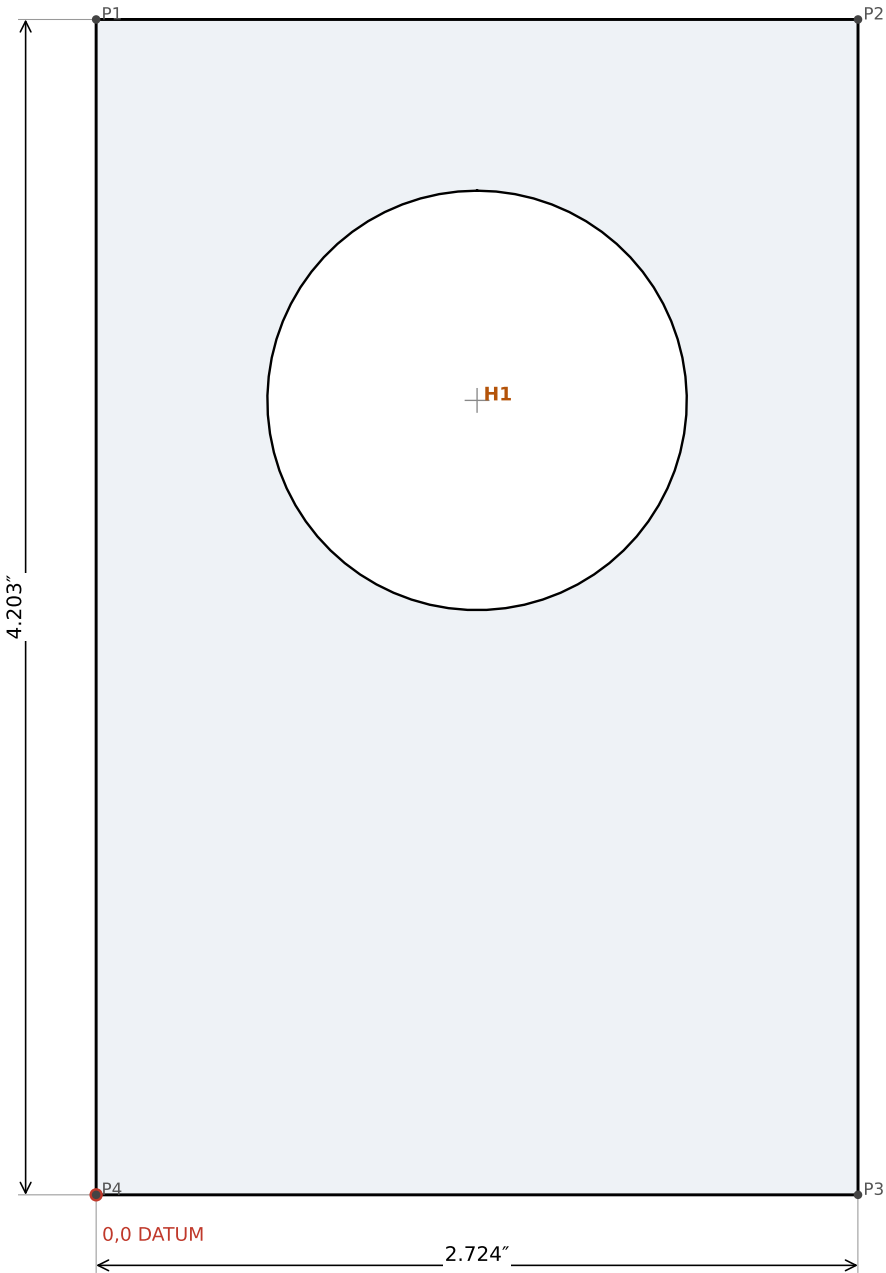
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-4 (welded version) — blank 4 of 8	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Slanted plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 4 / 8

FR-4 (welded version) — blank 5 of 8

Bottom plate (hole Ø1.5 for the tube) · blank 2.724” × 4.203” · HR steel sheet 0.138” · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø1.500” X 1.362 Y 2.841

OUTLINE POINTS (X, Y from datum)

P1 0.000 4.203
P2 2.724 4.203
P3 2.724 0.000
P4 0.000 0.000

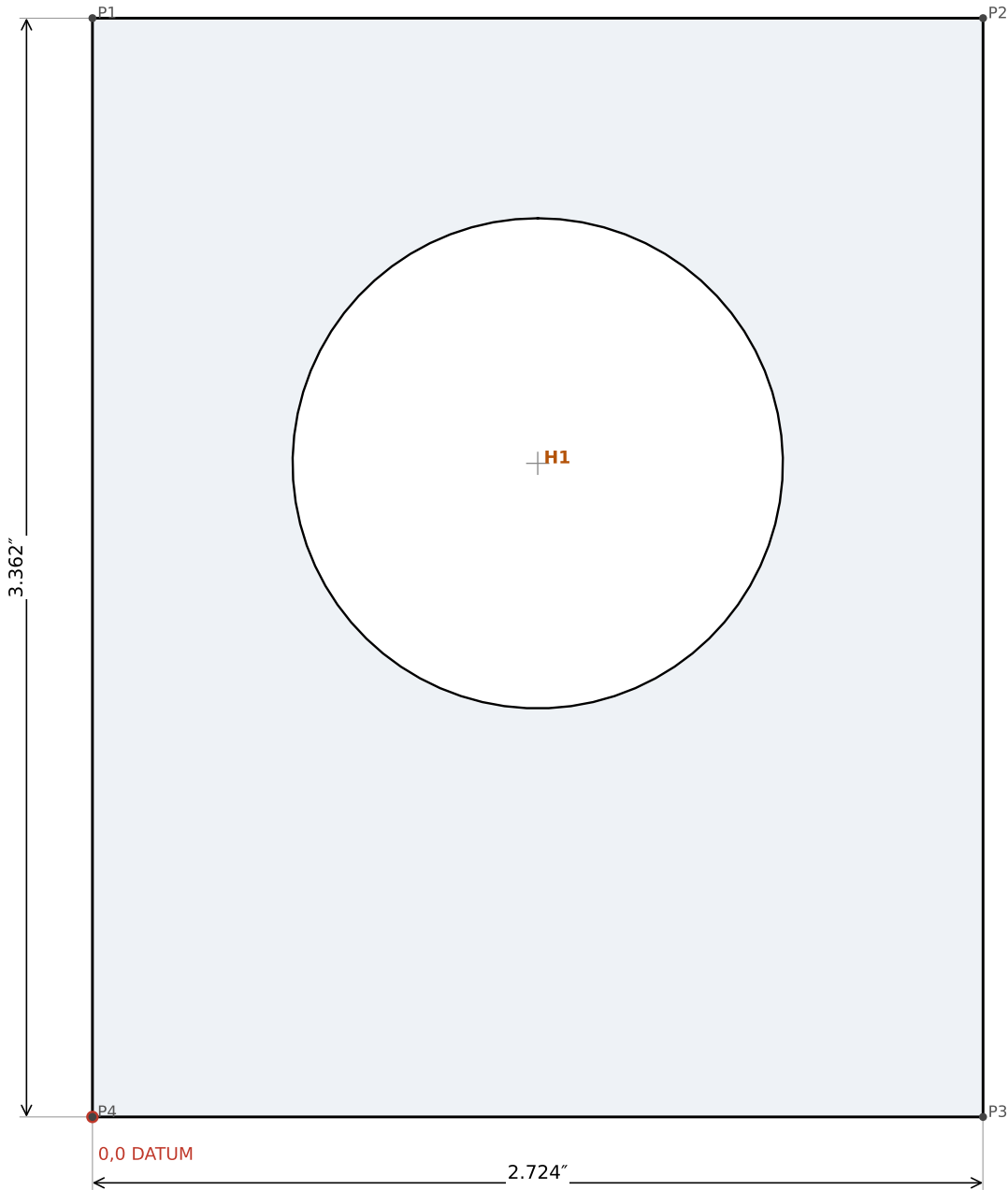
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138” (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * + tube OD1.5 / ID1.181 through (bottom + top), weld all round

FR-4 (welded version) — blank 5 of 8	HR steel sheet 0.138”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Bottom plate (hole Ø1.5 for the tube)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 5 / 8

FR-4 (welded version) — blank 6 of 8

Top plate (hole Ø1.5 for the tube) · blank 2.724” × 3.362” · HR steel sheet 0.138” · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø1.500” X 1.362 Y 2.000

OUTLINE POINTS (X, Y from datum)

P1 0.000 3.362
P2 2.724 3.362
P3 2.724 0.000
P4 0.000 0.000

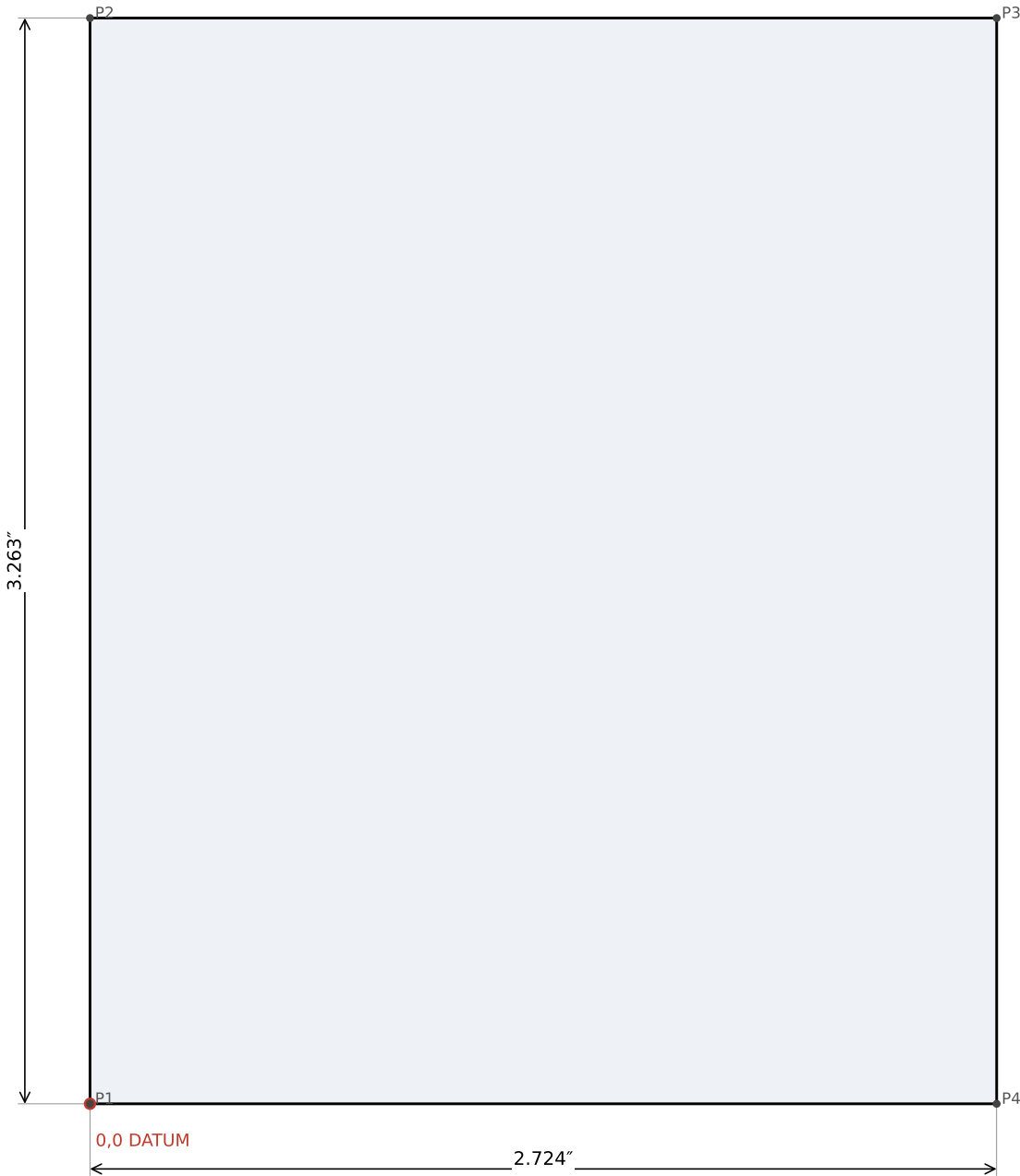
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138” (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * the tube passes through

FR-4 (welded version) — blank 6 of 8	HR steel sheet 0.138”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Top plate (hole Ø1.5 for the tube)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 6 / 8

FR-4 (welded version) — blank 7 of 8

Ear plate · blank 2.724" × 3.263" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	3.263
P3	2.724	3.263
P4	2.724	0.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-4 (welded version) — blank 7 of 8	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Ear plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 7 / 8

FR-4 (welded version) — blank 8 of 8

Ear cap (bend R0.625) · blank 2.705" × 1.200" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	2.705	0.000
P2	2.705	1.200
P3	0.000	1.200
P4	0.000	0.000



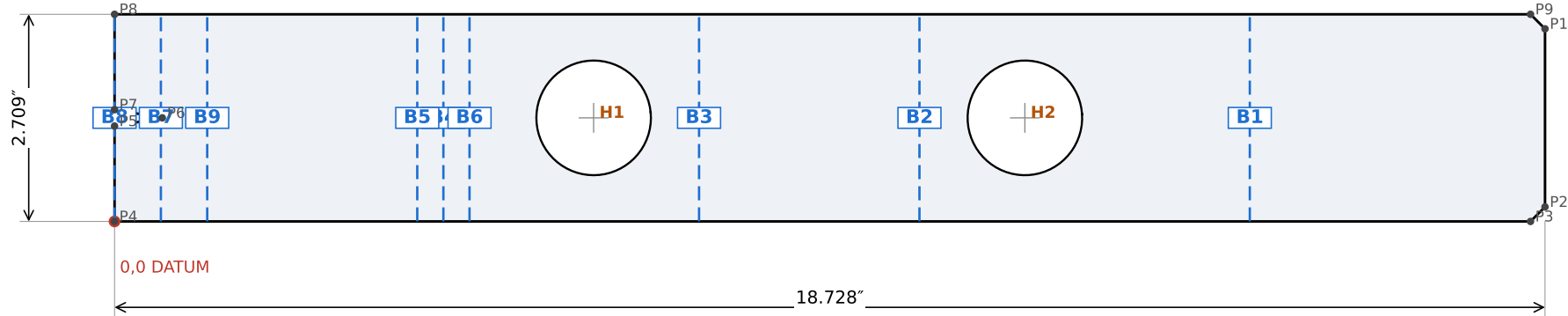
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * bend to arc R 0.625; V-notch 5/8

FR-4 (welded version) — blank 8 of 8	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Ear cap (bend R0.625)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 8 / 8

FR-4 (bent version) — blank 1 of 3

Wrap strip (ear, top, front, bottom, slant) · blank 18.728" × 2.709" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1	Ø1.500"	X 6.272	Y 1.355
H2	Ø1.500"	X 11.918	Y 1.355

BENDS (bend centre line, from datum) IR = t = 0.138"

B1	(5) 60°	(14.863, 0.000) → (14.863, 2.709)
B2	(4) 90°	(10.538, 0.000) → (10.538, 2.709)
B3	(3) 90°	(7.653, 0.000) → (7.653, 2.709)
B4	(2) 90° ↓ R0.38	(4.305, 0.000) → (4.305, 2.709)
B5	radius-bend edge	(3.963, 0.000) → (3.963, 2.709)
B6	radius-bend edge	(4.647, 0.000) → (4.647, 2.709)
B7	(1) 126.9° R0.49	(0.607, 0.000) → (0.607, 2.709)
B8	radius-bend edge	(0.000, 0.000) → (0.000, 2.709)
B9	radius-bend edge	(1.213, 0.000) → (1.213, 2.709)
radius-bend edge = start/end of a large-radius bend (bump bend between them)		

OUTLINE POINTS (X, Y from datum)

P1	18.728	2.519
P2	18.728	0.190
P3	18.538	0.000
P4	0.000	0.000
P5	0.000	1.248
P6	0.625	1.355
P7	0.000	1.462
P8	0.000	2.709
P9	18.538	2.709

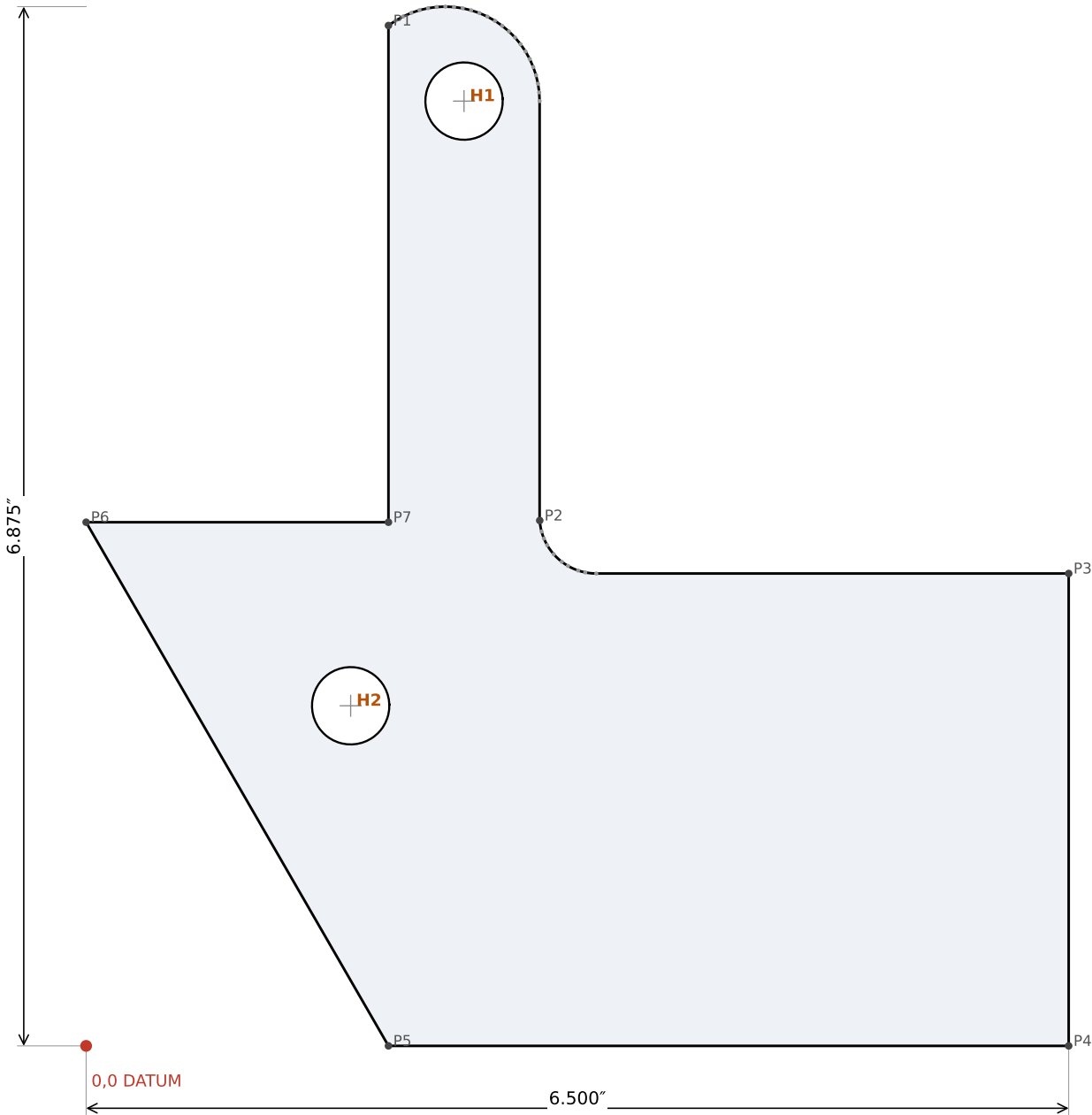
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).
- * arc C-D bent over R 0.48; R ⅜" at E; no corner reliefs

FR-4 (bent version) — blank 1 of 3	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Wrap strip (ear, top, front, bottom, slant)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 4

FR-4 (bent version) — blank 2 of 3

Side plate FR-4 (holes 0.512) · blank 6.500" × 6.875" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.512"	X 2.500	Y 6.250
H2	Ø0.512"	X 1.750	Y 2.250

OUTLINE POINTS (X, Y from datum)

P1	2.000	6.750
P2	3.001	3.475
P3	6.500	3.125
P4	6.500	0.000
P5	2.000	0.000
P6	0.000	3.464
P7	2.000	3.464

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.048	6.782
2.098	6.810
2.151	6.833
2.206	6.852
2.262	6.865
2.319	6.872
2.377	6.875
2.434	6.872
2.491	6.864
2.547	6.851
2.602	6.832
2.655	6.809
2.705	6.781
2.752	6.748
2.797	6.711
2.837	6.671
2.874	6.626
2.907	6.579
2.935	6.528
2.958	6.476
2.976	6.421
2.989	6.365
2.997	6.308
3.000	6.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.013	3.403
3.029	3.357
3.050	3.312
3.093	3.253
3.147	3.203
3.188	3.175
3.255	3.145
3.302	3.132
3.375	3.125

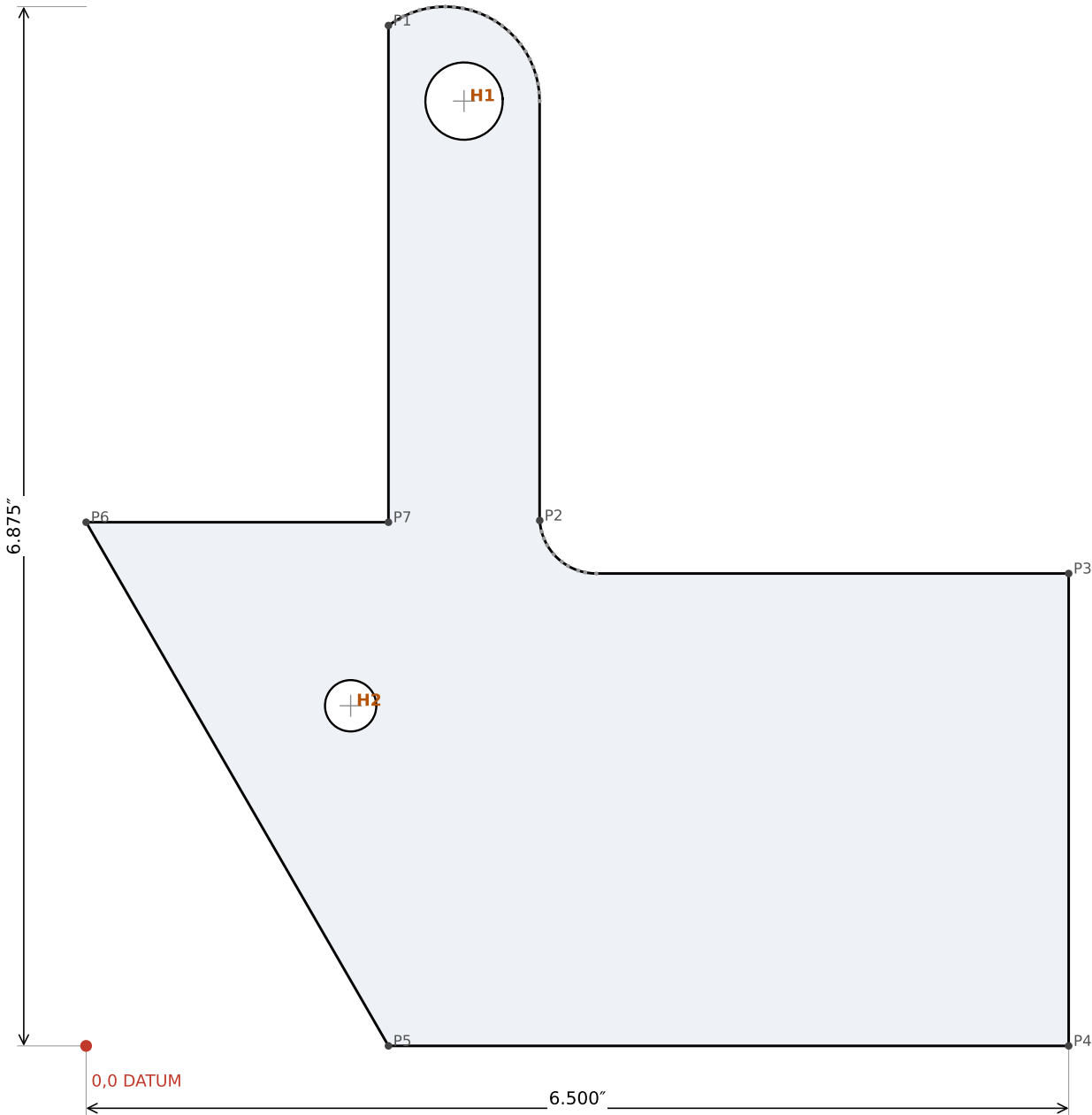
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * flat, no bends

FR-4 (bent version) — blank 2 of 3	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate FR-4 (holes 0.512)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 4

FR-4 (bent version) — blank 3 of 3

Side plate FL-4 (holes 0.512 and 0.34) · blank 6.500" × 6.875" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.512"	X 2.500	Y 6.250
H2	Ø0.340"	X 1.750	Y 2.250

OUTLINE POINTS (X, Y from datum)

P1	2.000	6.750
P2	3.001	3.475
P3	6.500	3.125
P4	6.500	0.000
P5	2.000	0.000
P6	0.000	3.464
P7	2.000	3.464

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.048	6.782
2.098	6.810
2.151	6.833
2.206	6.852
2.262	6.865
2.319	6.872
2.377	6.875
2.434	6.872
2.491	6.864
2.547	6.851
2.602	6.832
2.655	6.809
2.705	6.781
2.752	6.748
2.797	6.711
2.837	6.671
2.874	6.626
2.907	6.579
2.935	6.528
2.958	6.476
2.976	6.421
2.989	6.365
2.997	6.308
3.000	6.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.013	3.403
3.029	3.357
3.050	3.312
3.093	3.253
3.147	3.203
3.188	3.175
3.255	3.145
3.302	3.132
3.375	3.125

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * flat, no bends

FR-4 (bent version) — blank 3 of 3	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate FL-4 (holes 0.512 and 0.34)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 4

FR-4 (bent version) — cut list (tubes)

- **Tube OD 1.5 / ID 1.181 — cut 3.689” (through bottom and top)**

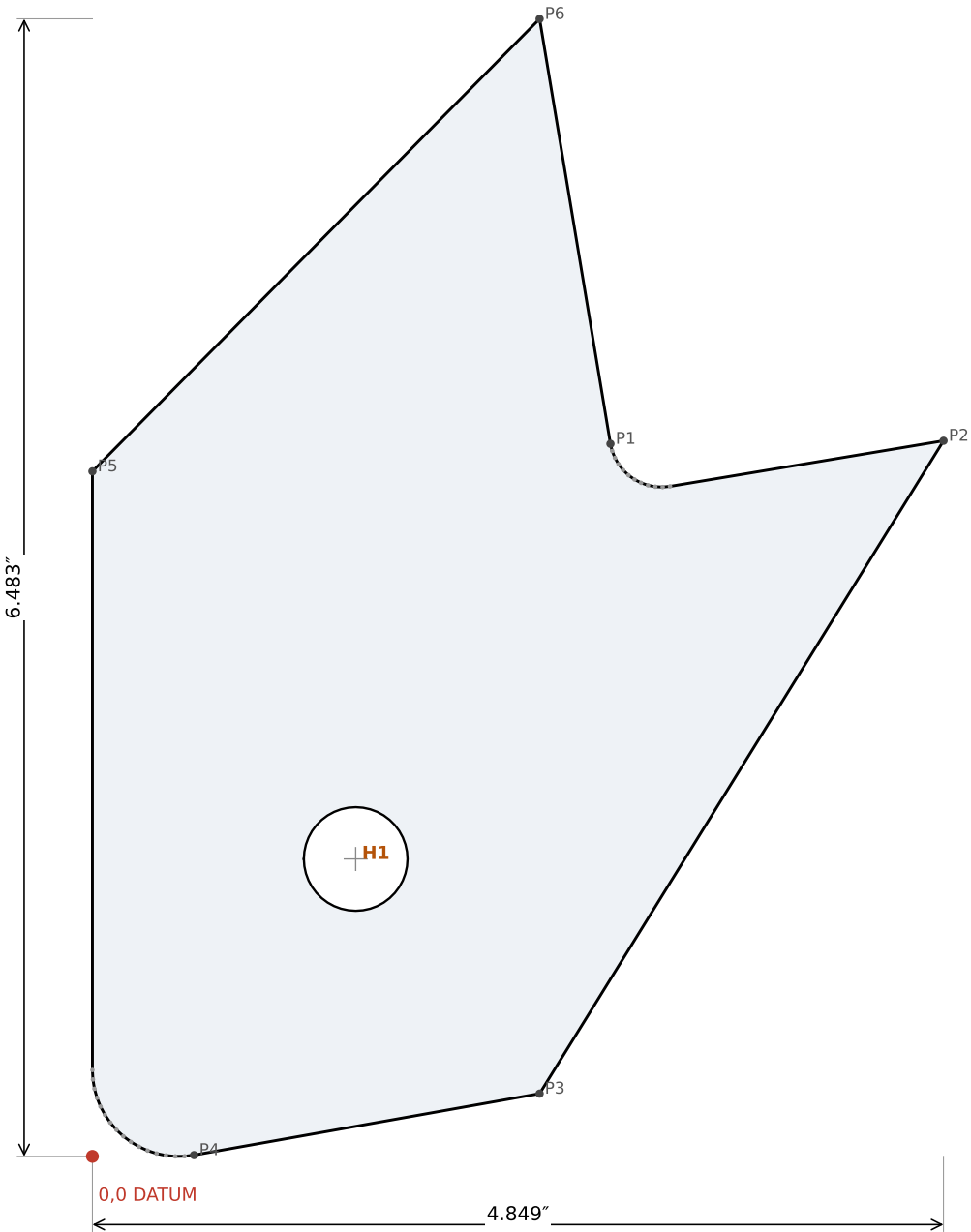
Stock: DOM round tube · cut length 3.689” · QTY 1 per side · FR + FL (mirror) = 2 per vehicle

Cut square, ±1/32”. Deburr. Lengths in inches.

FR-4 — tubes		inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Tube OD 1.5 / ID 1.181 — cut 3.689” (through bottom and top)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 4 / 4

FR-5 (welded version) — blank 1 of 2

Side plate · blank 4.849" × 6.483" · HR steel sheet 0.138" · 2 per side · FR + FL (mirror) = 4 per vehicle



HOLES (centre X, Y from datum)

H1 Ø0.591" X 1.500 Y 1.695

OUTLINE POINTS (X, Y from datum)

P1 2.951 4.061
P2 4.849 4.078
P3 2.548 0.358
P4 0.579 0.007
P5 0.000 3.904
P6 2.548 6.483

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.961 4.020
2.977 3.981
2.998 3.945
3.024 3.911
3.054 3.882
3.089 3.858
3.126 3.838
3.166 3.825
3.207 3.817
3.249 3.815
3.291 3.819

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

0.525 0.001
0.472 0.000
0.418 0.005
0.366 0.016
0.315 0.033
0.266 0.055
0.219 0.082
0.176 0.115
0.137 0.151
0.102 0.192
0.072 0.236
0.046 0.284
0.026 0.334
0.012 0.385
0.003 0.438
0.000 0.492

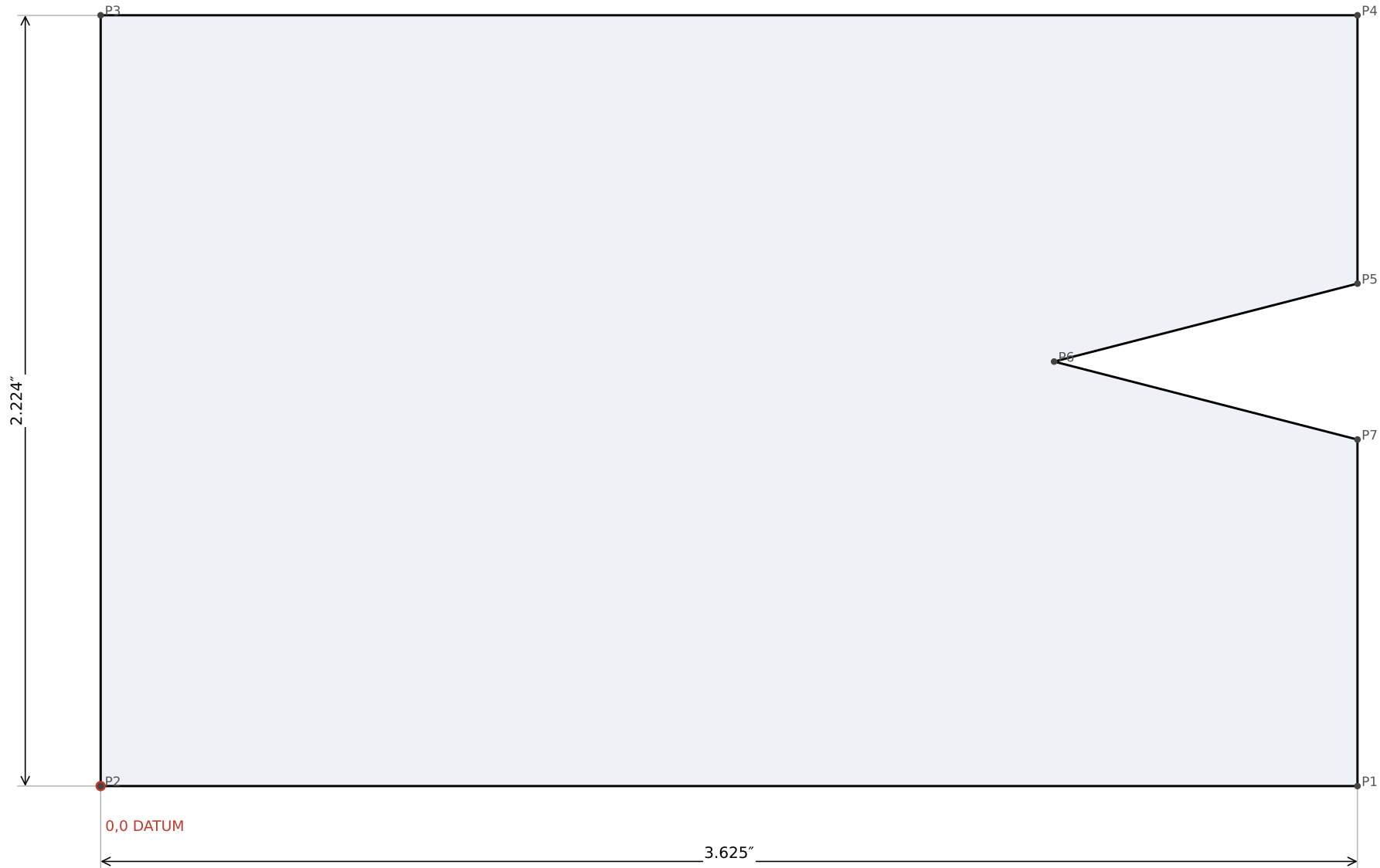
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-5 (welded version) — blank 1 of 2	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate	QTY: 2 per side · FR + FL (mirror) = 4 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 2

FR-5 (welded version) — blank 2 of 2

Plate on edge A-C · blank 3.625” × 2.224” · HR steel sheet 0.138” · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	3.625	0.000
P2	0.000	0.000
P3	0.000	2.224
P4	3.625	2.224
P5	3.625	1.450
P6	2.750	1.225
P7	3.625	1.000

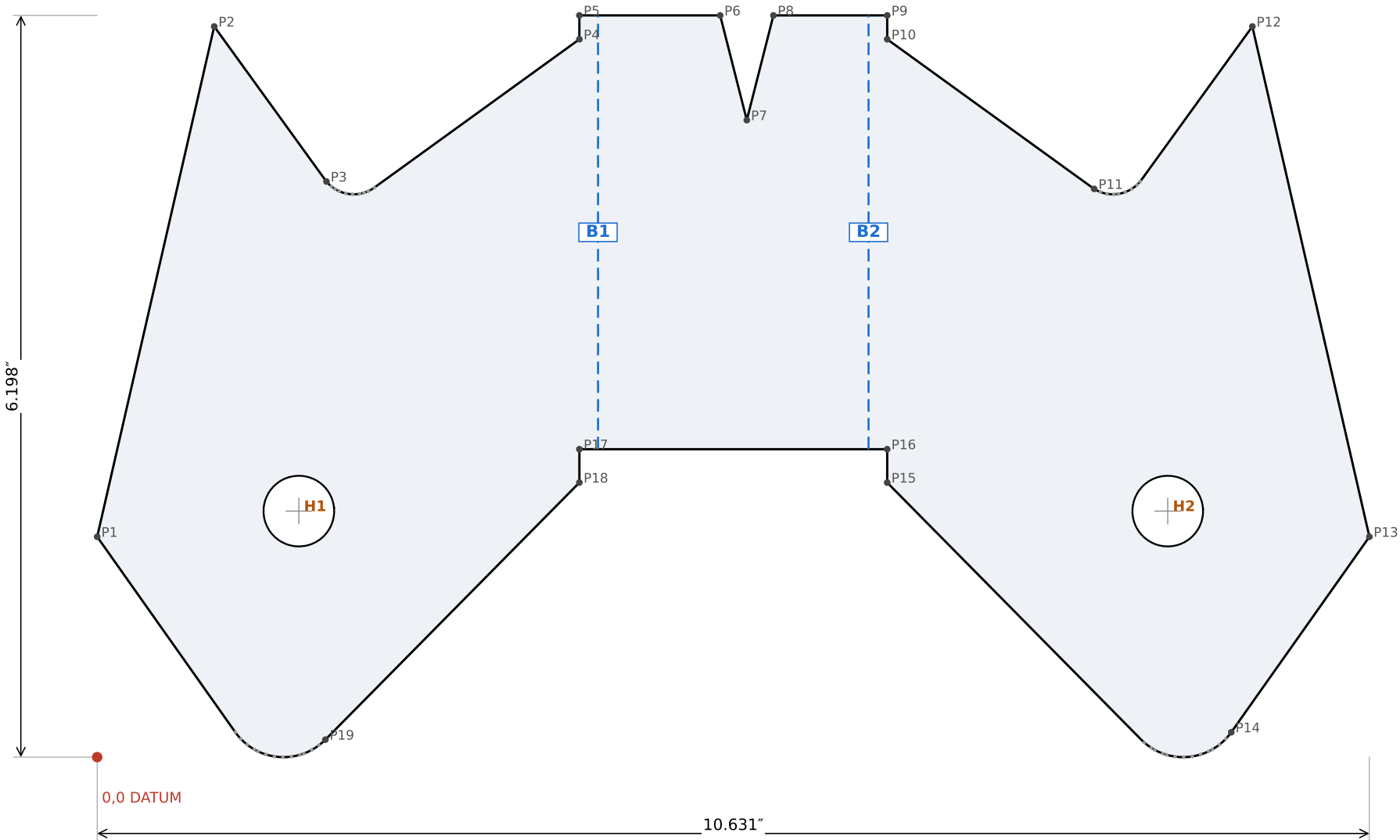
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138” (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-5 (welded version) — blank 2 of 2	HR steel sheet 0.138”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate on edge A-C	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 2 / 2

FR-5 (bent version) — blank 1 of 1

U-body (2 sides + plate A-C) · blank 10.631” × 6.198” · HR steel sheet 0.138” · 1 per side · FR + FL (mirror) = 2 per vehicle



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138” (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).
- * 90° bends, R = t

HOLES (centre X, Y from datum)

H1 Ø0.591” X 1.685 Y 2.056
H2 Ø0.591” X 8.946 Y 2.056

BENDS (bend centre line, from datum) IR = t = 0.138”

B1 (1) 90° (4.185, 2.573) → (4.185, 6.198)
B2 (2) 90° (6.446, 2.573) → (6.446, 6.198)

OUTLINE POINTS (X, Y from datum)

P1 0.000 1.842
P2 0.978 6.106
P3 1.916 4.810
P4 4.029 5.999
P5 4.029 6.198
P6 5.206 6.198
P7 5.428 5.323
P8 5.651 6.198
P9 6.602 6.198
P10 6.602 5.999
P11 8.331 4.749
P12 9.653 6.106
P13 10.631 1.842
P14 9.477 0.208
P15 6.602 2.295
P16 6.602 2.573
P17 4.029 2.573
P18 4.029 2.295
P19 1.907 0.146

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

1.957 4.770
1.988 4.747
2.022 4.729
2.077 4.711
2.135 4.704
2.192 4.708
2.230 4.717
2.266 4.730
2.316 4.760

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

8.383 4.723
8.439 4.708
8.496 4.704
8.535 4.707
8.591 4.722
8.643 4.747
8.689 4.782
8.715 4.810

FREE-FORM CURVE 3 (grey dots, follow DXF) — points X, Y

9.431 0.153
9.378 0.105
9.319 0.065
9.287 0.048
9.220 0.022
9.151 0.006
9.079 0.000
9.008 0.004
8.938 0.019
8.904 0.031
8.838 0.060
8.808 0.079
8.751 0.122

FREE-FORM CURVE 4 (grey dots, follow DXF) — points X, Y

1.853 0.099
1.793 0.060

FR-5 (bent version) — blank 1 of 1	HR steel sheet 0.138”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
U-body (2 sides + plate A-C)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 1

FR-5 (bent version) — blank 1 of 1 — tables (continued)

1.728 0.031
1.694 0.019
1.623 0.004
1.552 0.000
1.480 0.006
1.411 0.022

1.344 0.048
1.312 0.065
1.253 0.105
1.200 0.153
1.154 0.208

FR-5 (bent version) — blank 1 of 1	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
U-body (2 sides + plate A-C)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

FR-6 — cut list (tubes)

• Tube 3×3 — cut 14 ¼”

Stock: Square tube 3” × 3”, wall 0.197” · cut length 14.250” · QTY 1 per side · FR + FL (mirror) = 2 per vehicle

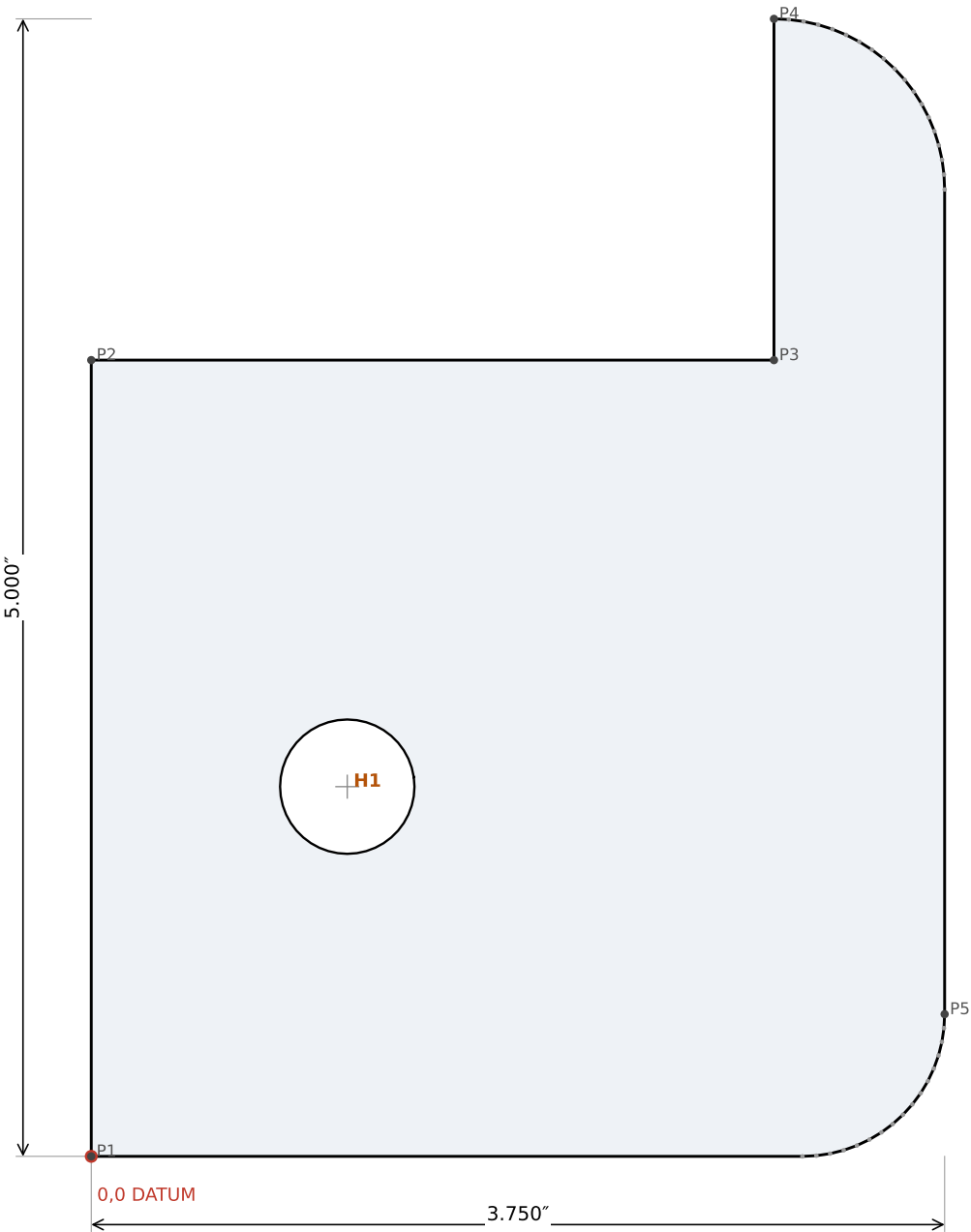
wall 0.197”

Cut square, ±1/32”. Deburr. Lengths in inches.

FR-6 — tubes		inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Tube 3×3 — cut 14 ¼”	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 1

FR-7 (welded version) — blank 1 of 3

Side plate with tab · blank 3.750″ × 5.000″ · HR steel sheet 0.138″ · 2 per side · FR + FL (mirror) = 4 per vehicle



HOLES (centre X, Y from datum)

H1 Ø0.591″ X 1.125 Y 1.625

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	3.500
P3	3.000	3.500
P4	3.000	5.000
P5	3.750	0.625

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

3.065	4.997
3.130	4.989
3.194	4.974
3.256	4.955
3.317	4.930
3.375	4.899
3.430	4.864
3.482	4.824
3.530	4.780
3.575	4.732
3.614	4.680
3.650	4.625
3.680	4.567
3.705	4.506
3.724	4.444
3.739	4.380
3.747	4.315
3.750	4.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.747	0.564
3.738	0.503
3.723	0.444
3.702	0.386
3.676	0.330
3.645	0.278
3.608	0.229
3.567	0.183
3.522	0.142
3.472	0.105
3.420	0.074
3.364	0.048
3.306	0.027
3.247	0.012
3.186	0.003
3.125	0.000

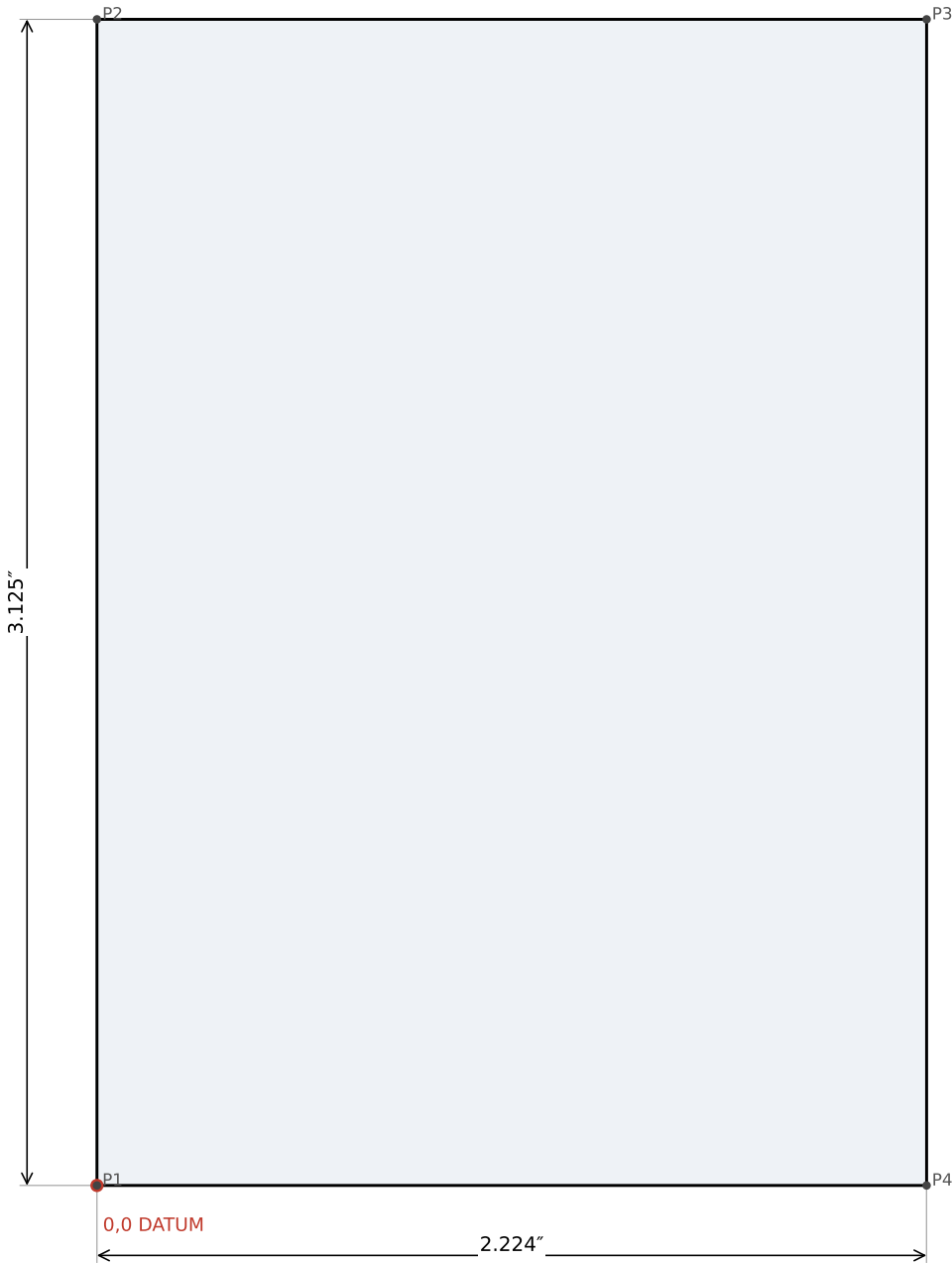
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138″ (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-7 (welded version) — blank 1 of 3	HR steel sheet 0.138″	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate with tab	QTY: 2 per side · FR + FL (mirror) = 4 per vehicle	cut ±0.010″ · bends ±1° · IR = t	SHEET 1 / 3

FR-7 (welded version) — blank 2 of 3

Bottom plate · blank 2.224" × 3.125" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	3.125
P3	2.224	3.125
P4	2.224	0.000

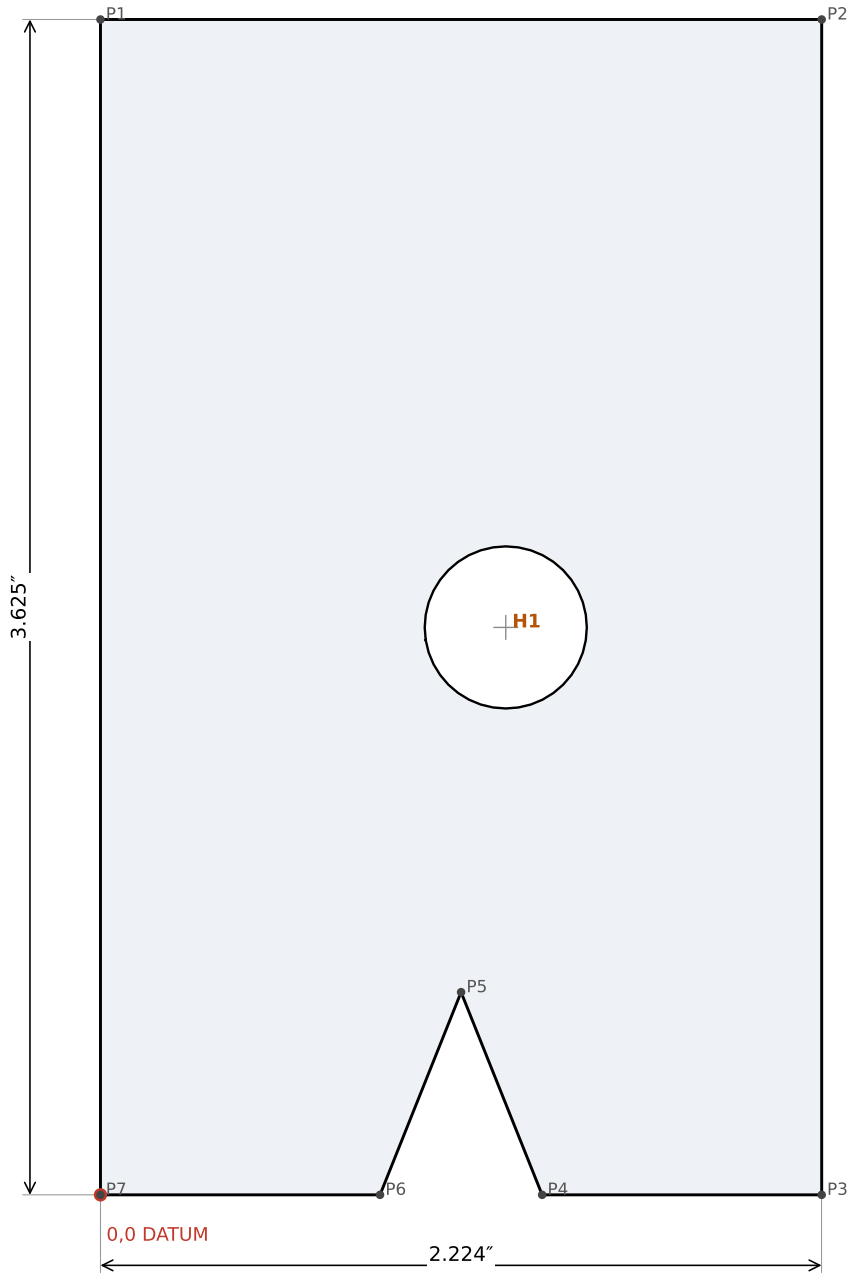
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-7 (welded version) — blank 2 of 3	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Bottom plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 3

FR-7 (welded version) — blank 3 of 3

Back plate (hole Ø0.5) · blank 2.224" × 3.625" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø0.500" X 1.250 Y 1.750

OUTLINE POINTS (X, Y from datum)

P1	0.000	3.625
P2	2.224	3.625
P3	2.224	0.000
P4	1.362	0.000
P5	1.112	0.625
P6	0.862	0.000
P7	0.000	0.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-7 (welded version) — blank 3 of 3	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Back plate (hole Ø0.5)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 3

FR-7 (bent version) — blank 1 of 3

L-strip (back + bottom) · blank 6.648" × 2.209" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)

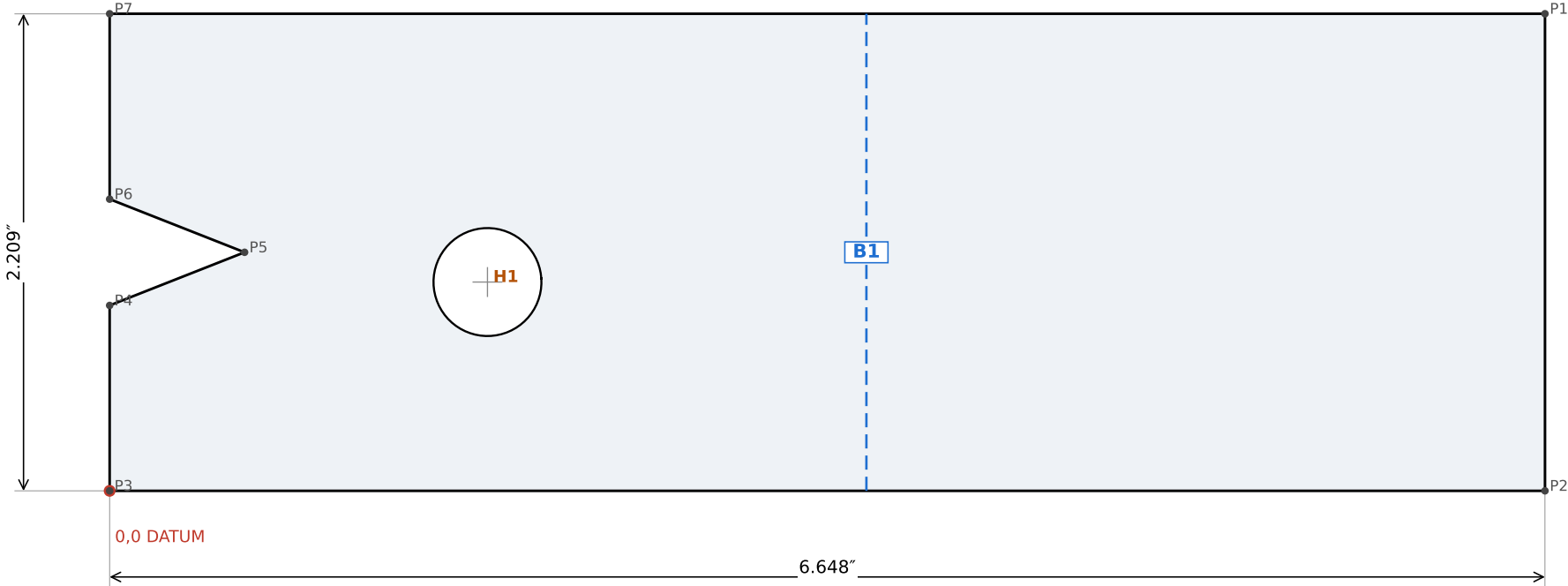
H1 Ø0.500" X 1.750 Y 0.967

BENDS (bend centre line, from datum) IR = t = 0.138"

B1 (1) 90° (3.505, 0.000) → (3.505, 2.209)

OUTLINE POINTS (X, Y from datum)

P1	6.648	2.209
P2	6.648	0.000
P3	0.000	0.000
P4	0.000	0.859
P5	0.625	1.105
P6	0.000	1.351
P7	0.000	2.209



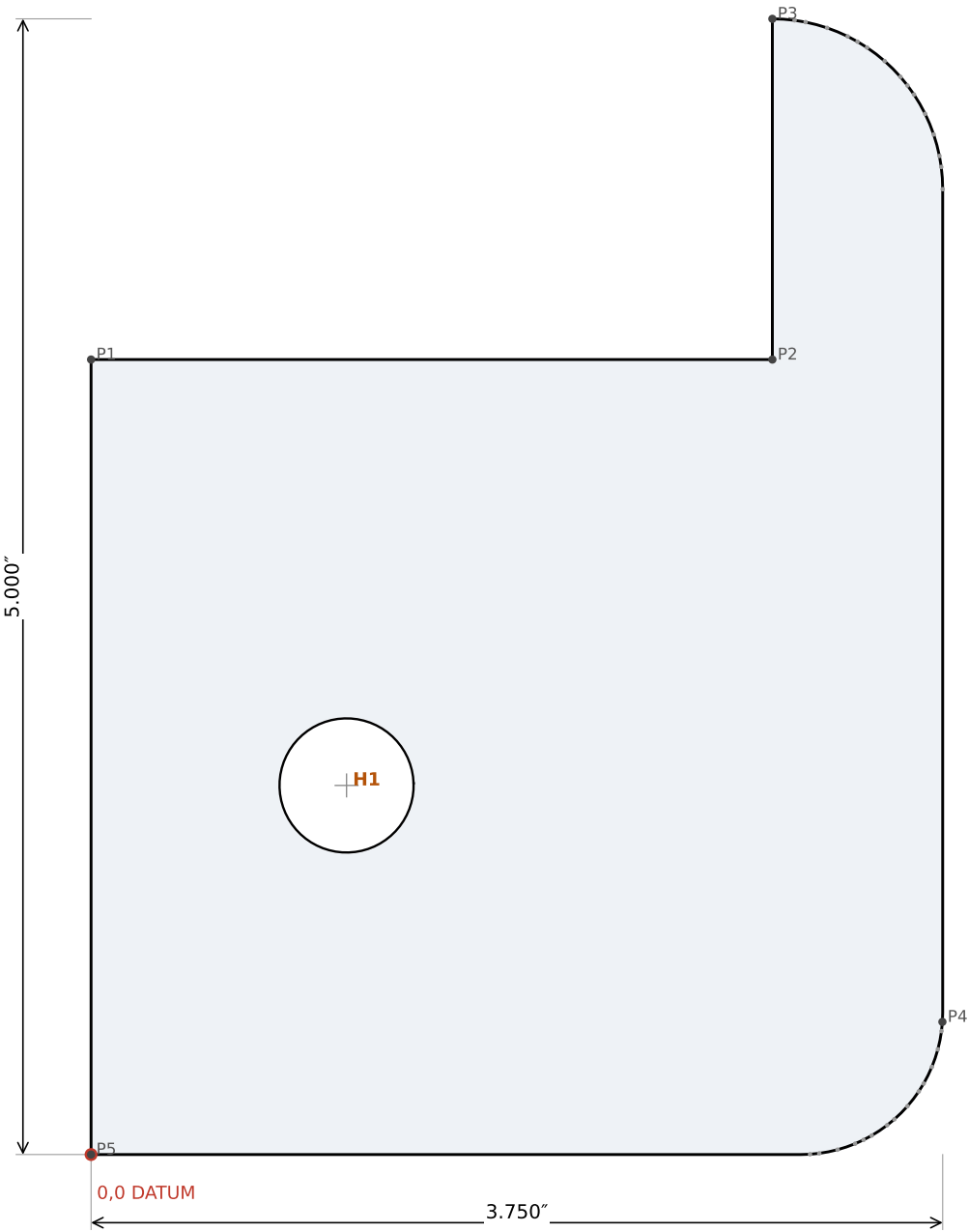
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * one 90° bend, R = t; V notch and Ø0.5 hole in the back

FR-7 (bent version) — blank 1 of 3	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
L-strip (back + bottom)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 3

FR-7 (bent version) — blank 2 of 3

Side plate · blank 3.750" × 5.000" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø0.591" X 1.125 Y 1.625

OUTLINE POINTS (X, Y from datum)

P1 0.000 3.500
P2 3.000 3.500
P3 3.000 5.000
P4 3.749 0.584
P5 0.000 0.000

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

3.098 4.994
3.146 4.986
3.241 4.960
3.332 4.923
3.375 4.899
3.417 4.874
3.494 4.814
3.564 4.745
3.595 4.707
3.624 4.667
3.673 4.582
3.710 4.491
3.736 4.396
3.744 4.348
3.750 4.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.745 0.543
3.729 0.463
3.702 0.386
3.666 0.312
3.645 0.278
3.595 0.213
3.537 0.155
3.506 0.129
3.438 0.084
3.401 0.065
3.364 0.048
3.287 0.021
3.207 0.005
3.166 0.001

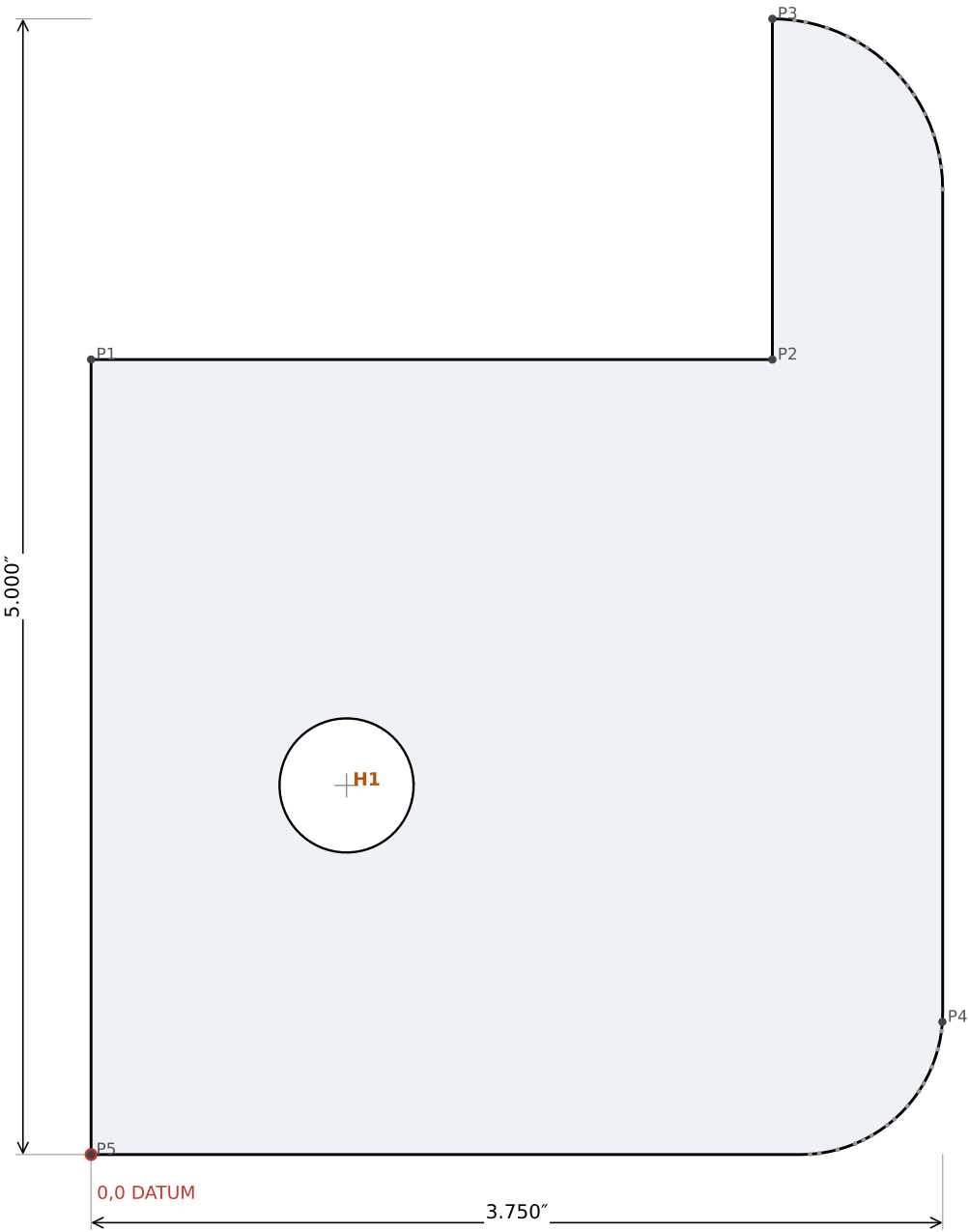
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * flat, no bends

FR-7 (bent version) — blank 2 of 3	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 3

FR-7 (bent version) — blank 3 of 3

Second side plate · blank 3.750" × 5.000" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø0.591" X 1.125 Y 1.625

OUTLINE POINTS (X, Y from datum)

P1 0.000 3.500
P2 3.000 3.500
P3 3.000 5.000
P4 3.749 0.584
P5 0.000 0.000

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

3.098 4.994
3.146 4.986
3.241 4.960
3.332 4.923
3.375 4.899
3.417 4.874
3.494 4.814
3.564 4.745
3.595 4.707
3.624 4.667
3.673 4.582
3.710 4.491
3.736 4.396
3.744 4.348
3.750 4.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.745 0.543
3.729 0.463
3.702 0.386
3.666 0.312
3.645 0.278
3.595 0.213
3.537 0.155
3.506 0.129
3.438 0.084
3.401 0.065
3.364 0.048
3.287 0.021
3.207 0.005
3.166 0.001

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * flat, no bends

FR-7 (bent version) — blank 3 of 3	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Second side plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 3

FR-8 (welded version) — blank 1 of 2

Side plate with arc · blank 12.000" × 1.875" · HR steel sheet 0.118" · 2 per side · FR + FL (mirror) = 4 per vehicle

HOLES (centre X, Y from datum)

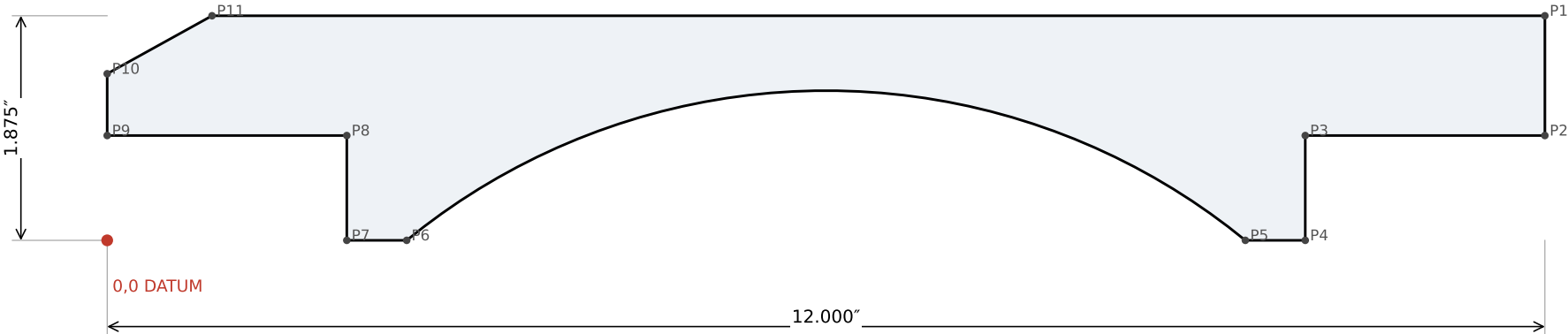
none

CURVED EDGES (radius, centre X, Y)

R1 R 5.525" centre 6.000, -4.275

OUTLINE POINTS (X, Y from datum)

P1	12.000	1.875
P2	12.000	0.875
P3	10.000	0.875
P4	10.000	0.000
P5	9.500	0.000
P6	2.500	0.000
P7	2.000	0.000
P8	2.000	0.875
P9	0.000	0.875
P10	0.000	1.391
P11	0.875	1.875



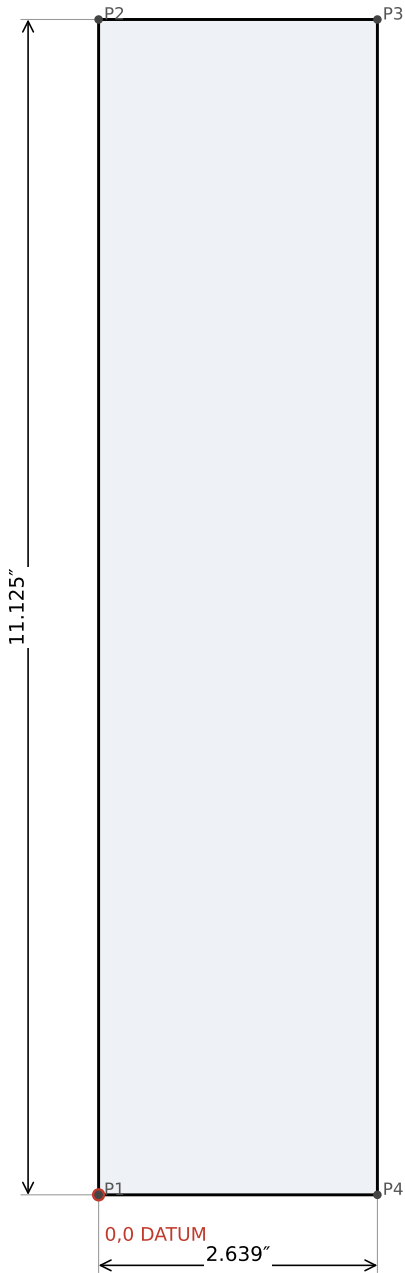
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.118" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-8 (welded version) — blank 1 of 2	HR steel sheet 0.118"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate with arc	QTY: 2 per side · FR + FL (mirror) = 4 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 2

FR-8 (welded version) — blank 2 of 2

Top plate · blank 2.639" × 11.125" · HR steel sheet 0.118" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	11.125
P3	2.639	11.125
P4	2.639	0.000

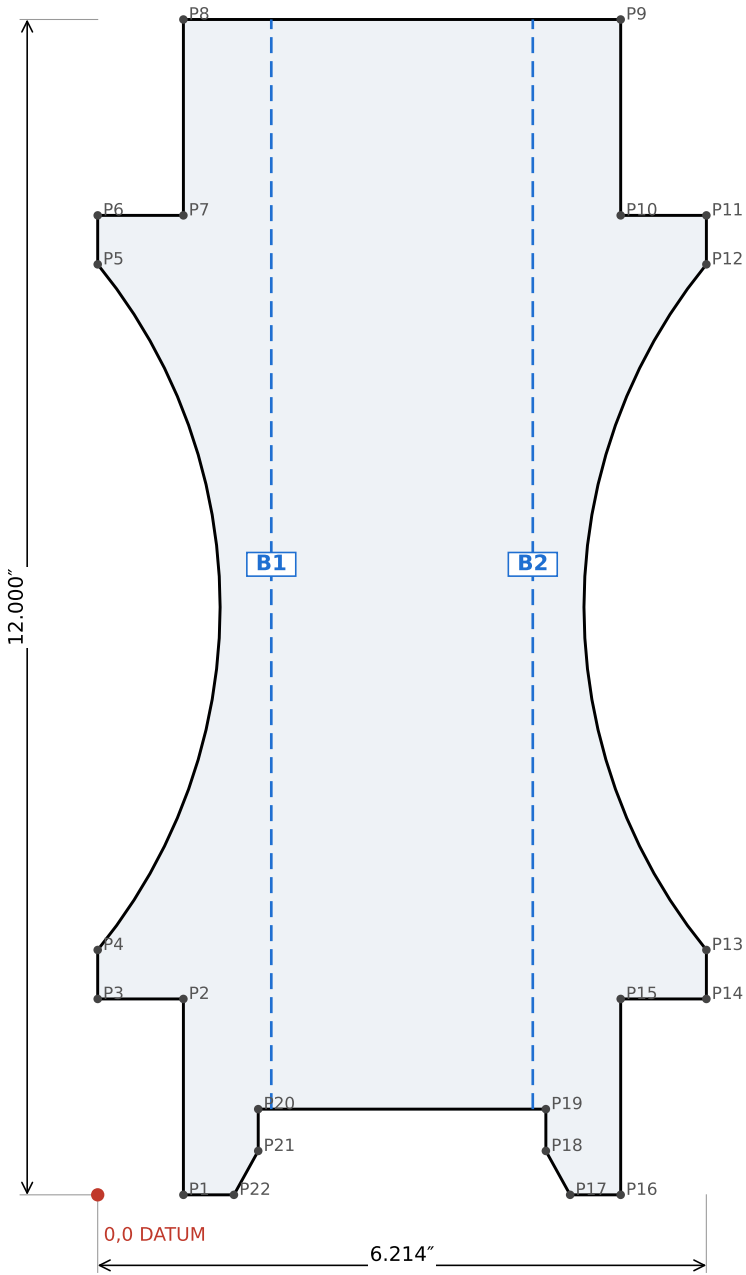
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.118" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-8 (welded version) — blank 2 of 2	HR steel sheet 0.118"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Top plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 2

FR-8 (bent version) — blank 1 of 1

Channel (top + 2 sides) · blank 6.214" × 12.000" · HR steel sheet 0.118" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

BENDS (bend centre line, from datum) IR = t = 0.118"

B1 (1) 90° (1.772, 0.875) → (1.772, 12.000)
B2 (2) 90° (4.442, 0.875) → (4.442, 12.000)

CURVED EDGES (radius, centre X, Y)

R1 R 5.525" centre -4.275, 6.000
R2 R 5.525" centre 10.489, 6.000

OUTLINE POINTS (X, Y from datum)

P1 0.875 0.000
P2 0.875 2.000
P3 0.000 2.000
P4 0.000 2.500
P5 0.000 9.500
P6 0.000 10.000
P7 0.875 10.000
P8 0.875 12.000
P9 5.340 12.000
P10 5.340 10.000
P11 6.215 10.000
P12 6.215 9.500
P13 6.215 2.500
P14 6.215 2.000
P15 5.340 2.000
P16 5.340 0.000
P17 4.824 0.000
P18 4.576 0.448
P19 4.576 0.875
P20 1.639 0.875
P21 1.639 0.448
P22 1.391 0.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.118" (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * 90° bends, R = t; relief cut at B

FR-8 (bent version) — blank 1 of 1	HR steel sheet 0.118"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Channel (top + 2 sides)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

FR-9 — cut list (tubes)

• Tube 3×3 — cut 8”

Stock: Square tube 3” × 3”, wall 0.197” · cut length 8.000” · QTY 1 per side · FR + FL (mirror) = 2 per vehicle

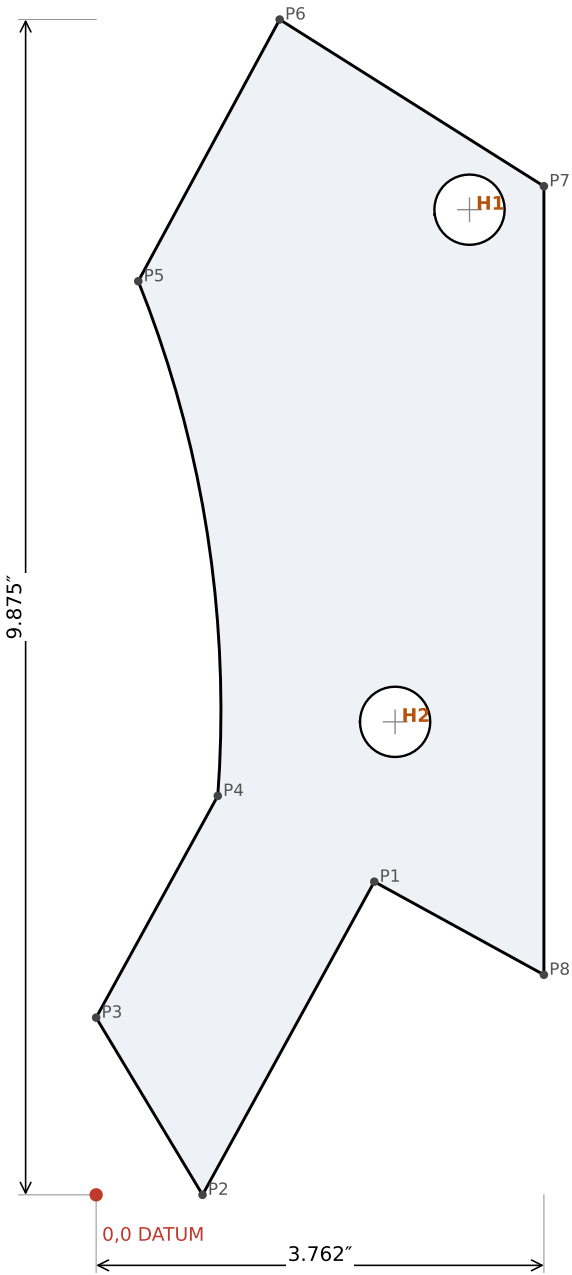
wall 0.197”

Cut square, ±1/32”. Deburr. Lengths in inches.

FR-9 — tubes		inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Tube 3×3 — cut 8”	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 1

FR-10 (welded version) — blank 1 of 7

Side plate (right + left) · blank 3.762” × 9.875” · HR steel sheet 0.138” · 2 per side · FR + FL (mirror) = 4 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.591"	X 3.137	Y 8.277
H2	Ø0.591"	X 2.512	Y 3.974

CURVED EDGES (radius, centre X, Y)

R1	R 9.700"	centre -8.651, 4.071
----	----------	----------------------

OUTLINE POINTS (X, Y from datum)

P1	2.337	2.631
P2	0.894	0.000
P3	0.000	1.489
P4	1.022	3.352
P5	0.354	7.676
P6	1.542	9.875
P7	3.762	8.474
P8	3.762	1.849

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138” (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-10 (welded version) — blank 1 of 7	HR steel sheet 0.138”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate (right + left)	QTY: 2 per side · FR + FL (mirror) = 4 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 7

FR-10 (welded version) — blank 2 of 7

Plate A-H · blank 6.625" × 2.724" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	2.724
P2	6.625	2.724
P3	6.625	0.000
P4	0.000	0.000



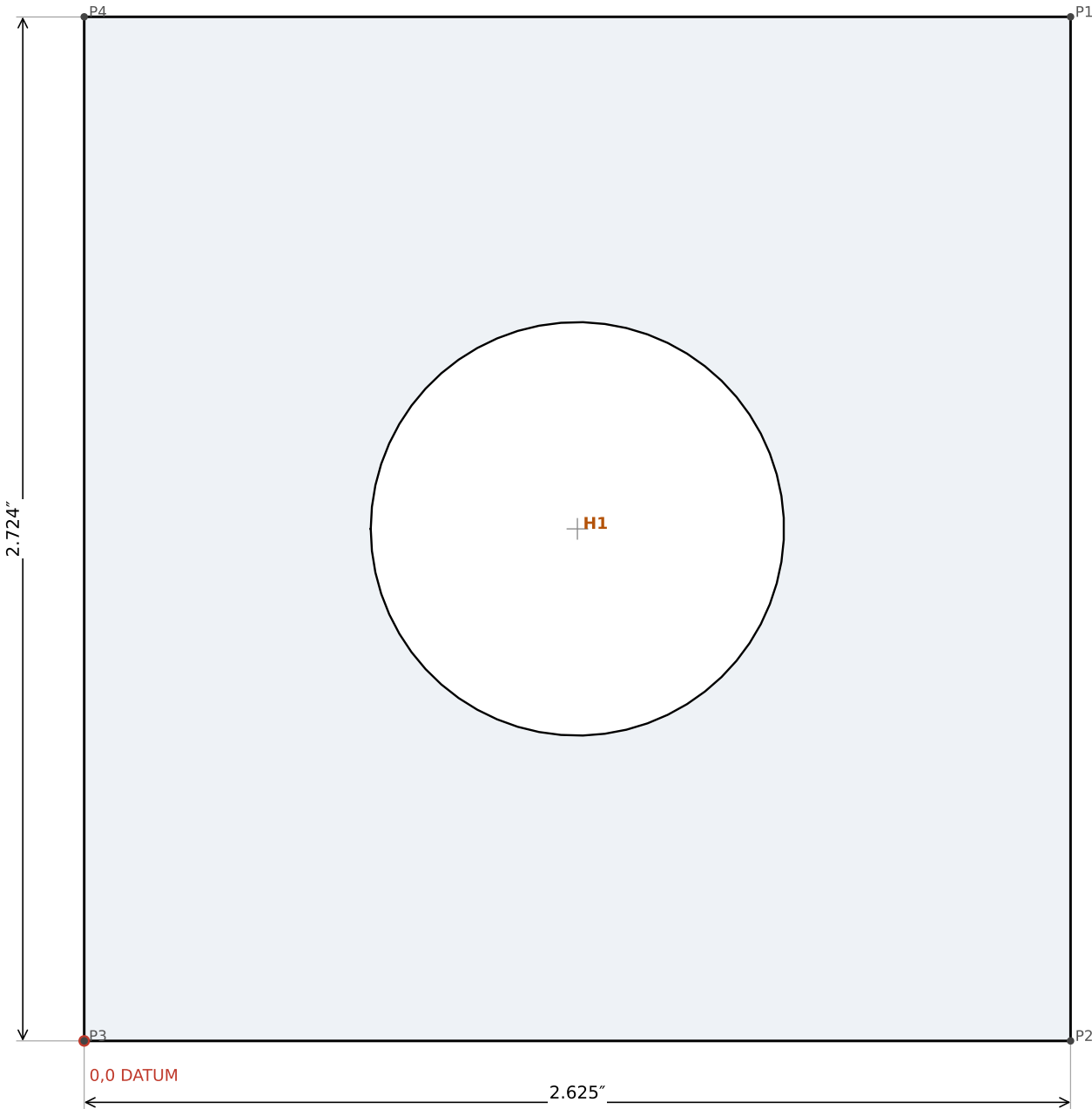
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-10 (welded version) — blank 2 of 7	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate A-H	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 7

FR-10 (welded version) — blank 3 of 7

Plate A-B (hole Ø1.1) · blank 2.625" × 2.724" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1	Ø1.100"	X 1.313	Y 1.362
----	---------	---------	---------

OUTLINE POINTS (X, Y from datum)

P1	2.625	2.724
P2	2.625	0.000
P3	0.000	0.000
P4	0.000	2.724

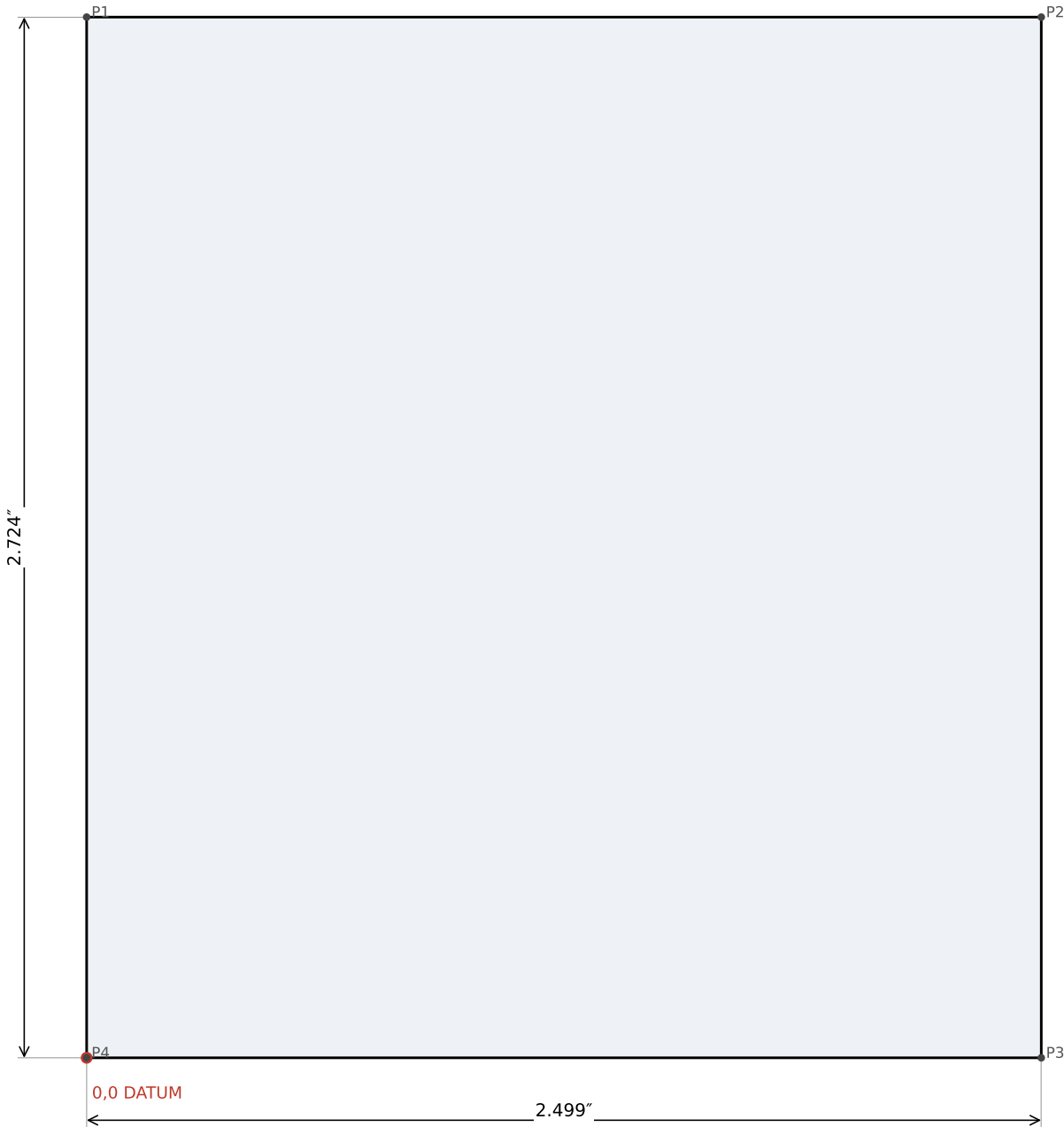
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * + bushing OD1.5×0.4

FR-10 (welded version) — blank 3 of 7	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate A-B (hole Ø1.1)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 7

FR-10 (welded version) — blank 4 of 7

Plate B-C · blank 2.499" × 2.724" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	2.724
P2	2.499	2.724
P3	2.499	0.000
P4	0.000	0.000

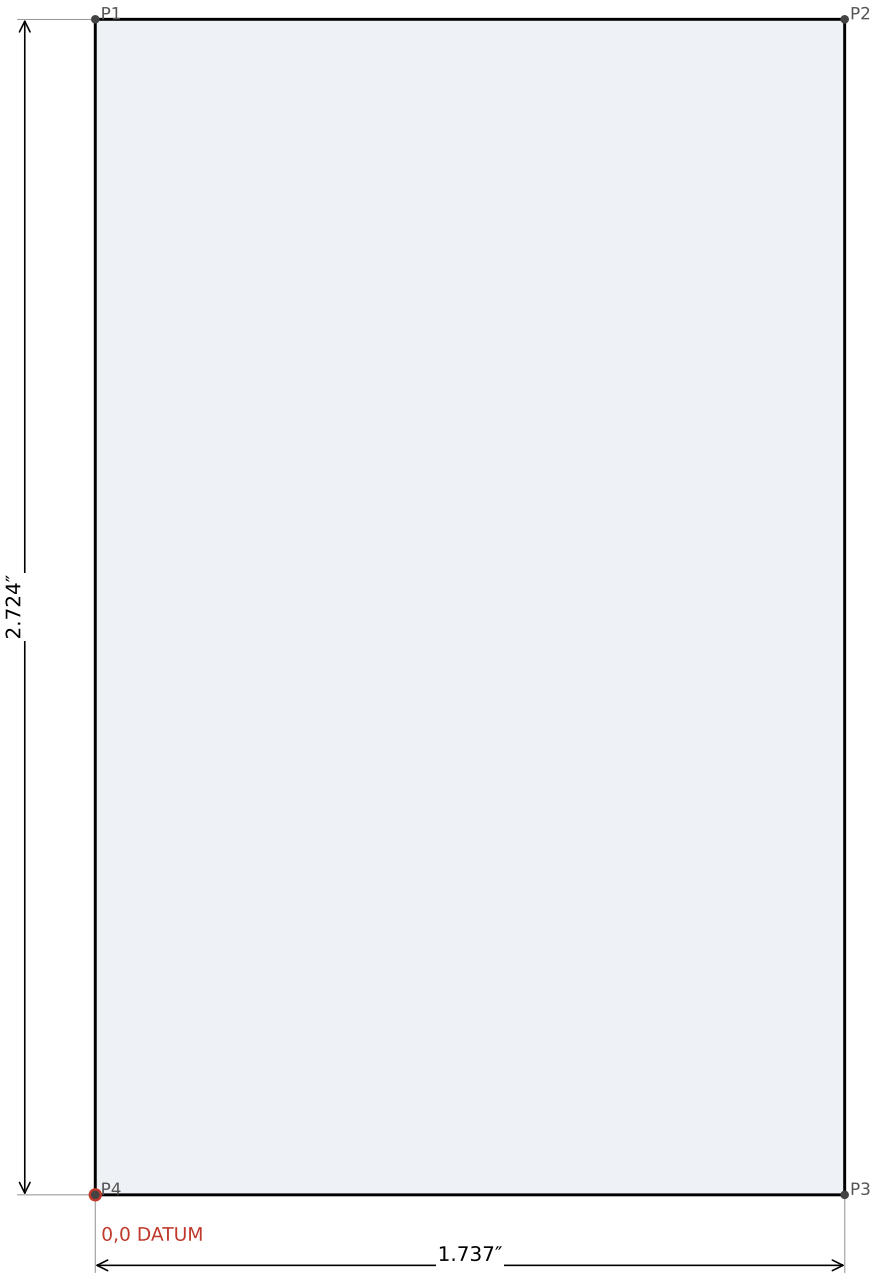
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-10 (welded version) — blank 4 of 7	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate B-C	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 4 / 7

FR-10 (welded version) — blank 5 of 7

Plate E-F · blank 1.737" × 2.724" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	2.724
P2	1.737	2.724
P3	1.737	0.000
P4	0.000	0.000

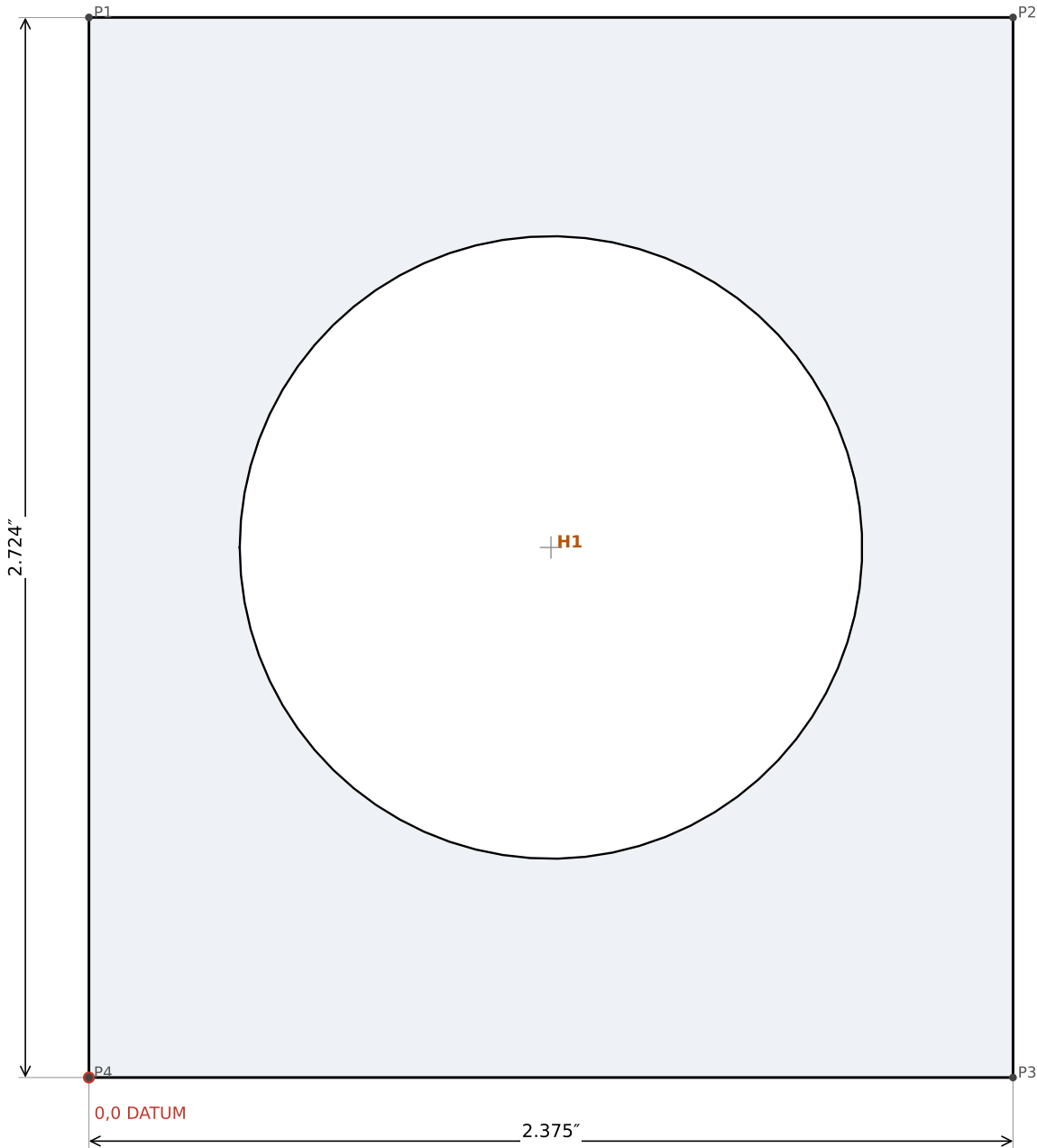
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-10 (welded version) — blank 5 of 7	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate E-F	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 5 / 7

FR-10 (welded version) — blank 6 of 7

Shelf (hole Ø1.181 for the bushing) · blank 2.375" × 2.724" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø1.600" X 1.187 Y 1.362

OUTLINE POINTS (X, Y from datum)

P1 0.000 2.724
P2 2.375 2.724
P3 2.375 0.000
P4 0.000 0.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * + bushing OD1.6×0.394

FR-10 (welded version) — blank 6 of 7	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Shelf (hole Ø1.181 for the bushing)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 6 / 7

FR-10 (welded version) — blank 7 of 7

Inner plate · blank 5.748” × 2.724” · HR steel sheet 0.138” · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	2.724
P3	5.748	2.724
P4	5.748	0.000



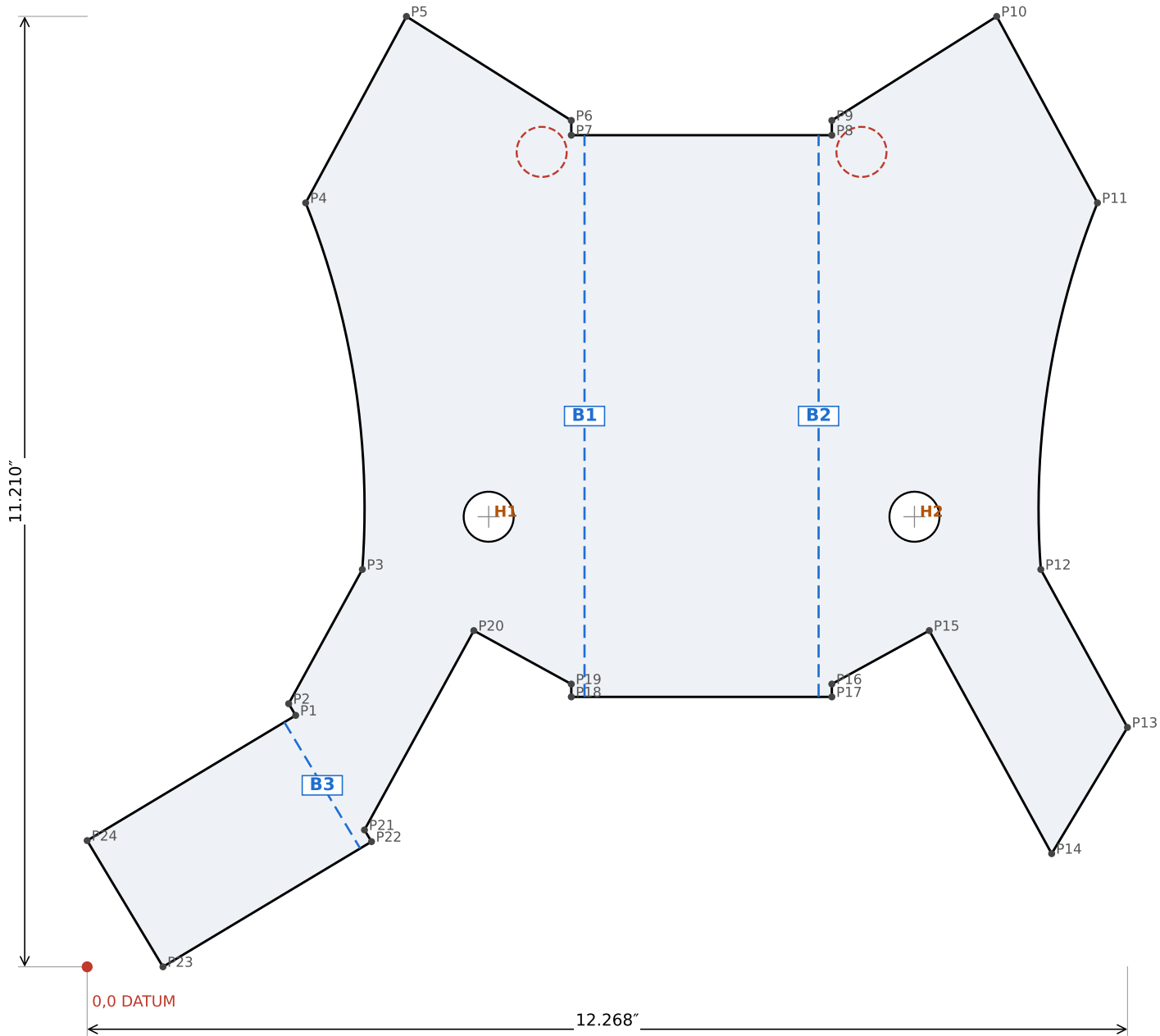
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138” (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-10 (welded version) — blank 7 of 7	HR steel sheet 0.138”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Inner plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 7 / 7

FR-10 (bent version) — blank 1 of 2

Part 1 "U-body" — 2 sides + wall A-H + flange E-F (3 bends) · blank 12.268" × 11.210" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.591"	X 4.735	Y 5.309
H2	Ø0.591"	X 9.756	Y 5.309

DRILL AFTER WELDING (red dashed — do NOT laser cut)

D1	Ø0.591"	X 5.360	Y 9.612
D2	Ø0.591"	X 9.131	Y 9.612

BENDS (bend centre line, from datum) IR = t = 0.138"

B1	(2) 90°	(5.865, 3.184) → (5.865, 9.809)
B2	(3) 90°	(8.626, 3.184) → (8.626, 9.809)
B3	(1) 90°	(2.325, 2.885) → (3.220, 1.396)

CURVED EDGES (radius, centre X, Y)

R1	R 9.699"	centre -6.427, 5.406
R2	R 9.700"	centre 20.919, 5.406

OUTLINE POINTS (X, Y from datum)

P1	2.459	2.966
P2	2.376	3.103
P3	3.245	4.687
P4	2.577	9.011
P5	3.765	11.210
P6	5.709	9.982
P7	5.709	9.809
P8	8.781	9.809
P9	8.781	9.982
P10	10.726	11.210
P11	11.914	9.011
P12	11.246	4.687
P13	12.268	2.824
P14	11.374	1.335
P15	9.931	3.966
P16	8.781	3.335
P17	8.781	3.184
P18	5.709	3.184
P19	5.709	3.335
P20	4.560	3.966
P21	3.270	1.615
P22	3.353	1.476
P23	0.894	0.000
P24	0.000	1.489

NOTES

1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 6. FR side as drawn; FL side = mirror image (flip the blank).
- * bends (1)(2)(3) 90° up, R = t; the 2 Ø0.591 holes near A (½" from A-B, ⅝" from A-H) are not in the flat — drill after bending

FR-10 (bent version) — blank 1 of 2	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Part 1 "U-body" — 2 sides + wall A-H + flange E-F (3 bends)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 3

FR-10 (bent version) — blank 2 of 2

Part 2 "strip" — A-B · B-C · shelf · inner plate (3 bends) · blank 12.460" × 2.709" · HR steel sheet 0.138" · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)

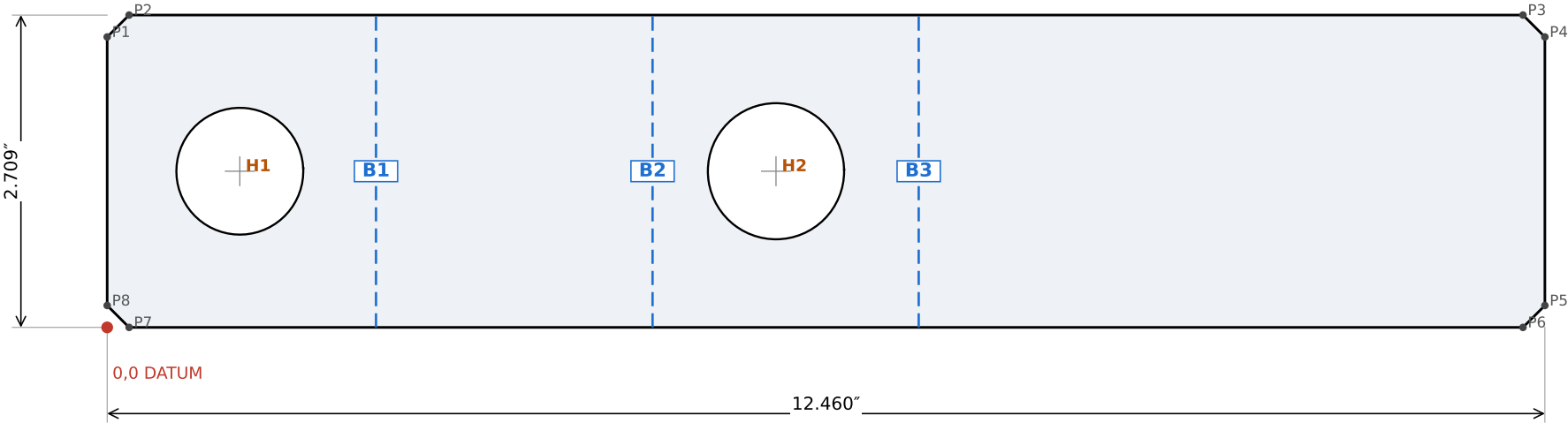
H1	Ø1.100"	X 1.150	Y 1.355
H2	Ø1.181"	X 5.796	Y 1.355

BENDS (bend centre line, from datum) IR = t = 0.138"

B1	(5) B 93.9°	(2.330, 0.000) → (2.330, 2.709)
B2	(6) C 86.1°	(4.727, 0.000) → (4.727, 2.709)
B3	(4) X 79.2° ↓	(7.033, 0.000) → (7.033, 2.709)

OUTLINE POINTS (X, Y from datum)

P1	0.000	2.519
P2	0.190	2.709
P3	12.270	2.709
P4	12.460	2.519
P5	12.460	0.190
P6	12.270	0.000
P7	0.190	0.000
P8	0.000	0.190



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.138" (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).
- * (4) X 79.2° opposite way; (5) B 93.9°, (6) C 86.1° up; corners 3/16"x45°

FR-10 (bent version) — blank 2 of 2	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Part 2 "strip" — A-B · B-C · shelf · inner plate (3 bends)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 3

FR-10 (bent version) — cut list (tubes)

• Ring OD 1.5 / ID 1.1 — cut 0.4” (on A-B)

Stock: DOM round tube · cut length 0.400” · QTY 1 per side · FR + FL (mirror) = 2 per vehicle

• Ring OD 1.6 / ID 1.181 — cut 0.394” (under the shelf)

Stock: DOM round tube · cut length 0.394” · QTY 1 per side · FR + FL (mirror) = 2 per vehicle

Cut square, ±1/32”. Deburr. Lengths in inches.

FR-10 — tubes		inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Ring OD 1.5 / ID 1.1 — cut 0.4” (on A-B)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 3 / 3